Shadow Mask-Driven Multimodal Intrinsic Image Decomposition for Hyperspectral and LiDAR Data Fusion

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***Abstract*— Remote sensing (RS) modalities from multisensor platforms, including hyperspectral (HS) and light detection and ranging (LiDAR) data, have been garnering increasing attention in overcoming the lack of information diversity. However, the specific modality correlation is always weakened and neglected. By reducing the spectral uncertainty, HS intrinsic image decomposition (HIID) has been proven to be effective in enhancing the HS image quality. It is an ill-posed problem and is challenged to capture the inherent modality connection. This article proposes a shadow mask-driven multimodal intrinsic image decomposition (smMIID) for HS and LiDAR data fusion. Notably, smMIID creatively clarifies and activates the potential of the crossmodal information in RS modalities and builds on the strength of the LiDAR elevation information when constructing HIID constraints. Our motivation is to overcome the deficiency of information diversity and modality correlation in existing intrinsic image decomposition (IID)-based frameworks for better data fusion performance. Classic IID methods decompose the HS image into one reflectance component (RC) and one shading component (SC). There are three constraints in smMIID: 1) the fundamental constraint on RC and SC; 2) the HS-LiDAR hybrid gradient-based constraint on RC; and 3) the LiDAR gradient-based constraint on SC. A shadow mask is obtained and utilized to guarantee spectral consistency when combining neighboring samples. Experiments on HS and LiDAR datasets prove that smMIID outperforms other techniques in terms of visualization and classification. The analyses are provided theoretically and experimentally to demonstrate that smMIID is robust to shadow mask generation.**

***Index Terms*— Classification, hyperspectral (HS) data, intrinsic image decomposition (IID), light detection and ranging (LiDAR) data, multimodality, remote sensing (RS).**

I. Introduction

REMOTE sensing (RS) is a detecting process of gathering the physical characteristics of on-ground objects by satellites and unmanned spacecraft from a distance [1]. The gathered information is always utilized for Earth observation tasks. Multiple sensor platforms have been applied to describe the same on-ground objects from diverse perspectives [2]. They systematically generate an immense number of RS modalities, including the panchromatic (PAN) image [3], multispectral (MS) image, hyperspectral (HS) image, light detection and ranging (LiDAR) data [4], etc. They reflect various distinct characteristics of targets and contain dissimilar peculiarities.

Earlier studies have verified that jointly using RS modalities could improve the classification performance [4]. Specifically, as a typical passive optical RS modality, the HS image contains abundant spectral-spatial information, making the HS classification tasks vibrant recently [5]. However, these tasks are always challenged due to the spectral redundancy, spatial mixture, and uneven illumination [6], [7]. As a typical active microwave RS modality, the LiDAR data provides detailed elevation information and has several elevation models [8], including the digital surface model (DSM), digital elevation model (DEM), etc. Existing modality fusion methods always align them by deep learning (DL) models [9], [10], neglecting their inherent connection. Hong et al. [11] presented three baselines (i.e., early fusion, middle fusion, and late fusion) and one multimodal DL baseline, which is a deep encoder-decoder network called EndNet. Zhang et al. [12] constructed an interleaving perception convolutional neural network (CNN) named IPCNN for integrating heterogeneous information. Zhang et al. [13] presented a structural optimization transmission framework for collaborative land-cover classification. Despite their notable performance, the existing methods ignore the inherent fusion mechanism and lack fusion interpretability. They always neglect the specific crossmodal information in HS and LiDAR data, hindering the improvement of the data fusion and classification performance. In this article, we aim to fully clarify and activate the HS-LiDAR inherent connection and highlight its contribution to HS and LiDAR data fusion.

Intrinsic image decomposition (IID) has attracted massive attention and played an increasingly important role when meeting challenging problems like object detection and scene identification [14]. IID attempts to decompose one red-green-blue (RGB) image into one reflectance component (RC) and one shading component (SC) [15], [16], [17]. They theoretically reflect the material properties of objects and environment

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illumination (EI), respectively. Fu et al. [18] proposed a novel method for high-quality intrinsic image (HQII) decomposition. Despite its excellent performance, directly using IID on HS images is unsuitable and infeasible. The reason is that the EI in HS images is highly relevant to the distant sunlight and atmosphere [19]. Therefore, the SC will be affected by the detailed wavelength, which is in contradiction to the assumption in IID. Kang et al. [20] firstly proposed a novel IID-based framework with three steps for HS feature extraction specifically. Inspired by this work, Jin et al. [21] improved it and proposed a novel reflectance model to recover intrinsic components from HS images. The authors designed two priors to reduce the uncertainties and improve the recovery ability. These methods contain multiple constraints on RC to solve the ill-posed problem. However, the constraints on SC are always unadopted and neglected in existing techniques, meaning that the SC with abundant geometric information is always undervalued. Furthermore, the HS and LiDAR crossmodal information can not be comprehensively captured and utilized. In this article, we attempt to exploit its potential and strength the inherent connection between HS and LiDAR data by introducing novel crossmodal constraints on RC and SC simultaneously.

HIID-based technologies assume that the RC has a strong local correlation, meaning that one RC sample can be represented by the linear combination of its spatial neighbors [19]. However, the EI in HS images is always severely affected by shadow, which is caused by land-cover, architecture, and vegetation [22]. The shadow makes the objects with the same material have diverse spectral characteristics in shadow and nonshadow regions. Combining the spatial neighbors in shadow regions to represent the sample in nonshadow regions tends to interfere with the decomposition process and obscure shadow edges in the RC. Furthermore, shadow detection and removal are challenging tasks and hard to obtain high accuracies. Thus, without removing shadow from HS images directly, this article analyzes the shadow interference and generates a shadow mask for decomposition to preserve distinct edges between shadow and nonshadow regions in components. This mask contributes to guaranteeing spectral consistency when combining neighboring samples. The robustness of the proposed method for shadow mask generation is analyzed theoretically and experimentally.

In this article, we propose a shadow mask-driven multimodal IID (smMIID) for HS and LiDAR data fusion to build on the strength of the LiDAR elevation information when constructing HIID constraints on RC and SC simultaneously. Our motivation is to overcome the deficiency of information diversity and RS crossmodal correlation in existing IID-based techniques. Different from other methods, smMIID utilizes the HS and LiDAR data simultaneously to constrain the crossmodal decomposition procedure, improving the specific fusion interpretability. Furthermore, the shadow mask generation in smMIID also guarantees spectral consistency when analyzing the local correlation in the decomposed components. Specifically, smMIID is based on three reasonable constraints: 1) the fundamental constraint on RC and SC; 2) the HS-LiDAR hybrid gradient-based constraint on RC; and 3) the LiDAR gradient-based constraint on SC. The first one considers the fundamental problem in HIID. The second one is presented based on both HS and LiDAR local gradients. It comprehensively draws on the strength of the shadow mask, which is generated by color space conversion and threshold filtering. This constraint aims to generate components with distinct edges between shadow and nonshadow regions. Assisted by the mask, the third constraint considers the LiDAR local gradient to restrict the SC optimization. The presented energy function is iteratively optimized by minimizing its Frobenius norm. A unique component initialization strategy (CIS) is presented for significant convergence performance. Considering that shadow detection and removal are challenging tasks, the robustness of smMIID to shadow mask generation is fully analyzed theoretically and experimentally. The main contributions of this article are summarized as follows.

1) A shadow mask-driven multimodal IID called smMIID is proposed, which can effectively exploit the inherent crossmodal connection between HS and LiDAR data. To the best of our knowledge, this is the first work to use the HS and LiDAR hybrid information to restrict the RC and SC optimization processes in HIID simultaneously.
2) Instead of removing shadow from HS images directly, a shadow mask is generated and introduced in smMIID to guarantee spectral consistency when analyzing the local correlation in RC and SC. With the assistance of the HS-LiDAR hybrid gradient, this strategy is fully implemented and contributes to preserving distinct edges between shadow and nonshadow regions in components.
3) The presented energy function is analyzed at length and optimized by minimizing its Frobenius norm. A unique component initialization strategy called CIS is introduced for significant convergence performance.

The remainder of this article is organized as follows. Section II provides some related works. Section III presents smMIID with three novel constraints in detail. Section IV reports the experimental results. Section V concludes this article.

II. Related Works

A. Intrinsic Image Decomposition

IID, under the Lambertian assumption, decomposes one RGB image $\mathcal{I}_{\mathcal{RGB}}$ into its components RC $\mathcal{R}_{\mathcal{IID}}$ and SC $\mathcal{S}_{\mathcal{IID}}$. The basic model of IID can be described as follows:

$$\mathcal{I}_{\mathcal{RGB}} = \mathcal{R}_{\mathcal{IID}} \odot \mathcal{S}_{\mathcal{IID}} \tag{1}$$

where $\odot$ is the element-wise multiplication operator. The shapes of $\mathcal{R}_{\mathcal{IID}}$ and $\mathcal{S}_{\mathcal{IID}}$ are the same with that of $\mathcal{I}_{\mathcal{RGB}}$. Apparently, (1) is ill-posed and cannot be optimized directly. Several methods have been proposed to solve this problem by considering additional constraints. Baslamisli et al. [23] proposed the reflectance and shading gradient descriptors for IID. Liu et al. [24] proposed an IID-based framework, directly extracting the latent features $\mathcal{R}_{\mathcal{IID}}$ and $\mathcal{S}_{\mathcal{IID}}$ from unsupervised and uncorrelated data. Zhang et al. [14] presented a nonlocal color compensation-based IID framework and chose the channels in the hue-saturation-value (HSV) color space as the complementary information. Zhang et al. [15]

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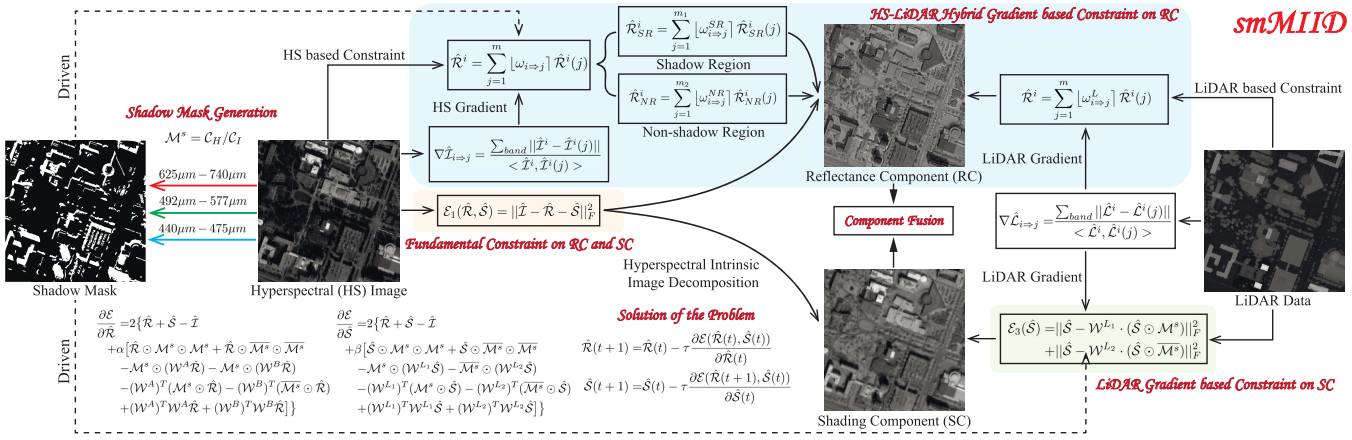


Fig. 1. Schematic of smMIID. The solid and dashed arrows represent the backbone and auxiliary information source of smMIID, respectively.

presented an unsupervised IID-based framework, which could achieve the decomposition effectively by self-similarity cues.

B. Hyperspectral IID

Different from the priori assumption in IID, the EI in the HS image $\mathcal{I}_{\mathcal{HS}}$ is associated with the distant sunlight and atmosphere [19]. Thus, the basic model of HIID, considering the detailed wavelength, can be described as follows:

$$\mathcal{I}_{\mathcal{HS}}(\lambda) = \mathcal{R}_{\mathcal{HIID}}(\lambda) \odot \mathcal{S}_{\mathcal{HIID}}(\lambda) \tag{2}$$

where λ is the wavelength. The difference between IID and HIID is that $\mathcal{S}_{\mathcal{IID}}$ does not vary with λ although $\mathcal{I}_{\mathcal{RGB}}$ also has three channels. Inspired by the unique HS characteristics, multiple HIID methods have been proposed. Jin et al. [21] proposed a HIID technique by introducing the reflectance and shading priors to reduce the HS uncertainties. Gu et al. [19] presented a novel HIID method called HyperIID to increase spatial resolution by fusing HS and PAN images. Gao et al. [25] optimized the existing multitemporal analyzing system and proposed a multitemporal IID to detect the land cover change situation. These methods prove that HIID plays a significant role in RS-related fields. However, they always consider the constraints on RC rather than SC, meaning that the SC with abundant geometric information tends to be undervalued and neglected. Furthermore, the potential of the generated SC, which is always removed directly, has not been fully exploited in land-cover classification tasks.

C. HS and LiDAR Data Fusion

Several studies have proven that HS and LiDAR data contain complementary information. Concretely, they contain spectral–spatial and elevation information, respectively. Hong et al. [11] proposed a multimodal DL baseline by following a deep encoder–decoder network to fuse and jointly classify HS and LiDAR data. Xu et al. [26] presented a CNN-based cascade network with two streams to extract useful RS information. Zhang et al. [12] developed an interleaving perception CNN (IPCNN) for integrating heterogeneous RS information and improving fusion and classification abilities. Hang et al. [27] solved the deficiency of information diversity and constructed an efficient and effective framework with coupled CNNs. On the whole, these techniques construct DL networks to comprehensively fuse RS modalities. Despite their great performance, they pay less attention to the specific RS inherent connection and lack fusion interpretability. Specifically, these techniques do not fully exploit the potential of HS and LiDAR data. In this article, we attempt to explore the inherent HS-LiDAR crossmodal correlation and highlight its contribution to increasing the performance of smMIID.

III. Proposed smMIID

We fuse the HS $\mathcal{I}_{\mathcal{HS}} \in \mathbb{R}^{n\times d}$ and LiDAR data $\mathcal{L} \in \mathbb{R}^{n\times 1}$ by jointly decomposing them into one RC and one SC initially, where n and d are the sample and band numbers, respectively. Subsequently, a simple but effective component fusion strategy is introduced. The detailed crossmodal constraints and energy functions are provided. The schematic of smMIID is shown in Fig. 1. Notably, the solid and dashed arrows represent the backbone and auxiliary information sources, respectively.

A. Shadow Mask Generation

In smMIID, the shadow mask is generated to increase the texture extraction abilities of samples near HS shadow edges. It contributes to obtaining intrinsic components with distinct edges between shadow and nonshadow regions. Considering that this generation procedure is not the research emphasis in this article, using a relatively complex method for extremely accurate results is unsuitable, which would interfere with technique evaluation. Thus, a conventional generation technique is chosen in smMIID [28]. Initially, the RGB bands are selected from $\mathcal{I}_{\mathcal{HS}}$ to reduce the computation time. Specifically, the red, green, and blue channels are chosen from the HS bands with the wavelengths of 625–740 μm, 492–577 μm and 440–475 μm, respectively. Subsequently, the combined image with three channels is converted to the hue-saturation-intensity (HSI) color space. Since the samples in shadow regions always have higher hue and lower intensity, the shadow mask $\mathcal{M}^s$ can be obtained as follows:

$$\mathcal{M}^s = \mathcal{C}_H/\mathcal{C}_I \tag{3}$$

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where $\mathcal{C}_H$ and $\mathcal{C}_I$ are the hue and intensity components in the HSI color space. The generated $\mathcal{M}^s$ is filtered by the OTSU thresholding strategy to detect the shadow regions. Specifically, the values in $\mathcal{M}^s$ are set as 1 when they are larger than the threshold θ. Otherwise, they are set as 0. Finally, some morphology methods are selected to revise the shadow mask, including: 1) removing spur samples; 2) removing tiny shadow regions; and 3) filling tiny holes. Notably, shadow detection and removal are challenging tasks, reducing the credibility of the detection results. Thus, without removing shadow from HS images directly, smMIID regards this shadow mask as auxiliary information for the whole decomposition procedure. It contributes to guaranteeing spectral consistency when combining neighboring samples. The robustness of smMIID to shadow mask generation is analyzed theoretically in Section III-G and verified experimentally in Section IV-E.

B. Fundamental Constraint on RC and SC

The basic model of HIID in (2) shows that it is ill-posed and intractable for optimization in terms of the $\odot$ operator. Thus, it is transformed into the logarithmic domain as follows:

$$\hat{\mathcal{I}}(\lambda) = \hat{\mathcal{R}}(\lambda) + \hat{\mathcal{S}}(\lambda) \tag{4}$$

where $\hat{\mathcal{I}} = \log(\mathcal{I}_{\mathcal{HS}}) \in \mathbb{R}^{n\times d}$, $\hat{\mathcal{R}} = \log(\mathcal{R}_{\mathcal{HIID}}) \in \mathbb{R}^{n\times d}$ and $\hat{\mathcal{S}} = \log(\mathcal{S}_{\mathcal{HIID}}) \in \mathbb{R}^{n\times d}$. Based on this model, the constraint $\mathcal{E}_1$ reflecting the relationship between $\hat{\mathcal{I}}$ and its components $\hat{\mathcal{R}}$ and $\hat{\mathcal{S}}$ is formulated as follows:

$$\mathcal{E}_1(\hat{\mathcal{R}}, \hat{\mathcal{S}}) = \|\hat{\mathcal{I}} - \hat{\mathcal{R}} - \hat{\mathcal{S}}\|_F^2 \tag{5}$$

where $\|\cdot\|_F^2$ is the Frobenius norm. This constraint guides the overall optimization direction of smMIID. However, it is difficult to be optimized for the global optimum solution. Considering this challenge, the following specific constraints, which consider the inherent HS and LiDAR connection, are presented for better convergence performance.

C. HS-LiDAR Hybrid Gradient-Based Constraint on RC

The HS image is one typical passive optical RS modality and reflects the spectral characteristics in the spectral domain. Generally, HIID assumes that one sample in $\hat{\mathcal{R}}$ is highly correlated with its spatial neighbors and can be combined by them linearly. The combination is formulated as follows:

$$\hat{\mathcal{R}}^i = \sum_{j=1}^{m} \lfloor \omega_{i\Rightarrow j} \rceil \hat{\mathcal{R}}^i(j) \tag{6}$$

where $\hat{\mathcal{R}}^i$ is one sample in $\hat{\mathcal{R}}$, $\hat{\mathcal{R}}^i(\cdot)$ is its spatial neighbor, m is the spatial neighbor number, $\omega_{i\Rightarrow j}$ denotes the correlation between $\hat{\mathcal{R}}^i$ and its jth neighbor, $\lfloor\cdot\rceil$ is the ipsative scaling operator. Considering that shadow tends to change the reflectance curves in optical RS modalities (i.e., $\hat{\mathcal{I}}$ and $\hat{\mathcal{R}}$), inevitable interference would appear if one sample is combined by its spatial neighbors in shadow regions and nonshadow regions simultaneously. Furthermore, the sample elevations in $\hat{\mathcal{L}}$ should also be considered when calculating the sample correlation. Thus, $\mathcal{M}^s$ is obtained in Section III-A and utilized for separating samples into different scenarios.

1) HS-Based Constraint in Shadow Regions: In $\mathcal{M}^s$, the values corresponding to the samples in shadow regions are denoted as 1. Thus, (6) can be rewritten for them as follows:

$$\hat{\mathcal{R}}^i_{\mathrm{SR}} = \sum_{j=1}^{m_1} \lfloor \omega^{\mathrm{SR}}_{i\Rightarrow j} \rceil \hat{\mathcal{R}}^i_{\mathrm{SR}}(j) \tag{7}$$

where $\hat{\mathcal{R}}^i_{\mathrm{SR}}$ is one $\hat{\mathcal{R}}$ sample in shadow regions, $\hat{\mathcal{R}}^i_{\mathrm{SR}}(\cdot)$ is its neighbor in shadow regions ($\hat{\mathcal{R}}^i_{\mathrm{SR}}$ has totally m neighbors, where m_1 neighbors lie in shadow regions) and $\omega^{\mathrm{SR}}_{i\Rightarrow j}$ is the correlation between $\hat{\mathcal{R}}^i_{\mathrm{SR}}$ and $\hat{\mathcal{R}}^i_{\mathrm{SR}}(j)$. The $\lfloor\cdot\rceil$ operator makes $\sum_{j=1}^{m_1} \omega^{\mathrm{SR}}_{i\Rightarrow j} = 1$. According to the Retinex theory [19], [29], $\hat{\mathcal{R}}$ is pixel-wise constant and consistent with $\hat{\mathcal{I}}$. The neighboring sample discontinuity in $\hat{\mathcal{I}}$ always results in weak correlations among these $\hat{\mathcal{R}}$ samples, corresponding to relatively low $\omega^{\mathrm{SR}}_{i\Rightarrow j}$ values. In smMIID, the sample continuity in $\hat{\mathcal{I}}$ is evaluated by the HS gradient $\nabla\hat{\mathcal{I}}_{i\Rightarrow j}$ as follows:

$$\nabla\hat{\mathcal{I}}_{i\Rightarrow j} = \frac{\sum_{\mathrm{band}} \|\hat{\mathcal{I}}^i - \hat{\mathcal{I}}^i(j)\|}{\langle \hat{\mathcal{I}}^i, \hat{\mathcal{I}}^i(j)\rangle} \tag{8}$$

where $\nabla\hat{\mathcal{I}}_{i\Rightarrow j}$ is the gradient between $\hat{\mathcal{I}}^i$ (corresponds to $\hat{\mathcal{R}}^i_{\mathrm{SR}}$) and its jth neighbor $\hat{\mathcal{I}}^i(j)$ (corresponds to $\hat{\mathcal{R}}^i_{\mathrm{SR}}(j)$), $\sum_{\mathrm{band}}$ is the sum of the deviations in all bands, $\|\cdot\|$ is the absolute operator, $\langle\cdot\rangle$ measures the spatial Euclidean distance. Thus, $\omega^{\mathrm{SR}}_{i\Rightarrow j}$ is formulated as follows:

$$\begin{aligned}
&① \ \text{if} \ \mathcal{M}^s(\hat{\mathcal{I}}^i) = 1 \ \text{and} \ \mathcal{M}^s(\hat{\mathcal{I}}^i(j)) = 1 \\
&\qquad \omega^{\mathrm{SR}}_{i\Rightarrow j} = \begin{cases} 1, & \|\nabla\hat{\mathcal{I}}_{i\Rightarrow j}\| \le \epsilon \\ \exp\left(-\dfrac{\|\nabla\hat{\mathcal{I}}_{i\Rightarrow j}\|^2}{\delta^2}\right), & \|\nabla\hat{\mathcal{I}}_{i\Rightarrow j}\| > \epsilon \end{cases} \\
&② \ \text{if} \ \mathcal{M}^s(\hat{\mathcal{I}}^i) = 0 \ \text{or} \ \mathcal{M}^s(\hat{\mathcal{I}}^i(j)) = 0 \\
&\qquad \omega^{\mathrm{SR}}_{i\Rightarrow j} = 0
\end{aligned} \tag{9}$$

where $\exp(\cdot)$ is the exponential operator, $\mathcal{M}^s(X)$ is the $\mathcal{M}^s$ value corresponding to X, $\delta = 1$ and $\epsilon = 10^{-5}$. Notably, if $\nabla\hat{\mathcal{I}}$ is small, the local structure in $\hat{\mathcal{R}}$ is preserved.

2) HS-Based Constraint in Nonshadow Regions: For the samples in nonshadow regions, we choose their nonshadow neighbors to combine them linearly. Thus, (6) can be rewritten as follows:

$$\hat{\mathcal{R}}^i_{\mathrm{NR}} = \sum_{j=1}^{m_2} \lfloor \omega^{\mathrm{NR}}_{i\Rightarrow j} \rceil \hat{\mathcal{R}}^i_{\mathrm{NR}}(j) \tag{10}$$

where $\hat{\mathcal{R}}^i_{\mathrm{NR}}$ is one $\hat{\mathcal{R}}$ sample in nonshadow regions, $\hat{\mathcal{R}}^i_{\mathrm{NR}}(\cdot)$ is its neighbor in nonshadow regions ($\hat{\mathcal{R}}^i_{\mathrm{NR}}$ has totally m neighbors, where m_2 neighbors lie in nonshadow regions). Thus, the correlation $\omega^{\mathrm{NR}}_{i\Rightarrow j}$ between $\hat{\mathcal{R}}^i_{\mathrm{NR}}$ and $\hat{\mathcal{R}}^i_{\mathrm{NR}}(j)$ is formulated as follows:

$$\begin{aligned}
&① \ \text{if} \ \overline{\mathcal{M}^s}(\hat{\mathcal{I}}^i) = 1 \ \text{and} \ \overline{\mathcal{M}^s}(\hat{\mathcal{I}}^i(j)) = 1 \\
&\qquad \omega^{\mathrm{NR}}_{i\Rightarrow j} = \begin{cases} 1, & \|\nabla\hat{\mathcal{I}}_{i\Rightarrow j}\| \le \epsilon \\ \exp\left(-\dfrac{\|\nabla\hat{\mathcal{I}}_{i\Rightarrow j}\|^2}{\delta^2}\right), & \|\nabla\hat{\mathcal{I}}_{i\Rightarrow j}\| > \epsilon \end{cases} \\
&② \ \text{if} \ \overline{\mathcal{M}^s}(\hat{\mathcal{I}}^i) = 0 \ \text{or} \ \overline{\mathcal{M}^s}(\hat{\mathcal{I}}^i(j)) = 0 \\
&\qquad \omega^{\mathrm{NR}}_{i\Rightarrow j} = 0
\end{aligned} \tag{11}$$

where $\overline{\mathcal{M}^s} = 1 - \mathcal{M}^s$ is the nonshadow mask.

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3) LiDAR-Based Constraint: When defining the sample correlation, only considering the local gradient in $\hat{\mathcal{I}}$ would be challenged by the information bias problem. Specifically, samples with different materials and similar reflectance curves could be chosen as spatial neighbors. Although they are not from the same category, their corresponding correlation is bigger than those of other sample couples. The LiDAR data in the logarithmic domain $\hat{\mathcal{L}} = \log(\mathcal{L})$, containing abundant elevation information and reflecting object geometric structures, is considered for solving this problem. In RS regions, targets with different elevations always have multiple materials and characteristics. Accordingly, we assume that the LiDAR gradient $\nabla\hat{\mathcal{L}}$ is related to the local sample correlation in the RC. If $\nabla\hat{\mathcal{L}}$ is small, the samples in local neighborhoods are continuous, meaning that the local structure in $\hat{\mathcal{R}}$ should be preserved. Formally, (6) can be rewritten as follows:

$$\begin{aligned}\hat{\mathcal{R}}^i &= \sum_{j=1}^{m} \lfloor \omega^L_{i\Rightarrow j} \rceil \hat{\mathcal{R}}^i(j) \\ \omega^L_{i\Rightarrow j} &= \begin{cases} 1, & \|\nabla\hat{\mathcal{L}}_{i\Rightarrow j}\| \le \epsilon \\ \exp\left(-\dfrac{\|\nabla\hat{\mathcal{L}}_{i\Rightarrow j}\|^2}{\delta^2}\right), & \|\nabla\hat{\mathcal{L}}_{i\Rightarrow j}\| > \epsilon \end{cases} \\ \nabla\hat{\mathcal{L}}_{i\Rightarrow j} &= \frac{\sum_{\text{band}} \|\hat{\mathcal{L}}^i - \hat{\mathcal{L}}^i(j)\|}{\langle \hat{\mathcal{L}}^i, \hat{\mathcal{L}}^i(j)\rangle} \end{aligned} \tag{12}$$

where $\nabla\hat{\mathcal{L}}_{i\Rightarrow j}$ is the gradient between $\hat{\mathcal{L}}^i$ (corresponding to $\hat{\mathcal{R}}^i$) and its jth neighbor $\hat{\mathcal{L}}^i(j)$ (corresponding to $\hat{\mathcal{R}}^i(j)$).

4) Aligned HS-LiDAR-Based Constraint: To simplify the constraints on RC, (7), (10) and (12) are aligned and combined as follows:

$$\begin{aligned}\hat{\mathcal{R}}^i = \sum_{j=1}^{m} \Big\{ &\lfloor \omega^{\text{SR}}_{i\Rightarrow j} + \omega^L_{i\Rightarrow j} \rceil \left[\hat{\mathcal{R}}^i(j) \odot \mathcal{M}^s(\hat{\mathcal{R}}^i(j))\right] \\ &+ \lfloor \omega^{\text{NR}}_{i\Rightarrow j} + \omega^L_{i\Rightarrow j} \rceil \left[\hat{\mathcal{R}}^i(j) \odot \overline{\mathcal{M}^s}(\hat{\mathcal{R}}^i(j))\right] \Big\}. \end{aligned} \tag{13}$$

Thus, the aligned HS-LiDAR hybrid gradient-based constraint on RC is formulated as follows:

$$\begin{aligned}\mathcal{E}_2(\hat{\mathcal{R}}) = &\|\hat{\mathcal{R}} - (\mathcal{W}^{\text{SR}} + \mathcal{W}^L) \cdot (\hat{\mathcal{R}} \odot \mathcal{M}^s)\|_F^2 \\ &+ \|\hat{\mathcal{R}} - (\mathcal{W}^{\text{NR}} + \mathcal{W}^L) \cdot (\hat{\mathcal{R}} \odot \overline{\mathcal{M}^s})\|_F^2 \end{aligned} \tag{14}$$

where $\mathcal{W}^* \in \mathbb{R}^{n\times n}$, $\mathcal{W}^*_{i\Rightarrow j} = \lfloor \omega^*_{i\Rightarrow j} \rceil$, $\mathcal{M}^s \in \mathbb{R}^{n\times d}$. If the ith sample x lies in shadow regions, $\mathcal{M}^s(x) = \mathcal{M}^s_{i,:} = 1$.

D. LiDAR Gradient-Based Constraint on SC

Existing HIID-based methods tend to consider multiple constraints on RC. However, the constraints on SC are always neglected. In smMIID, the geometric information in the SC is comprehensively considered for decomposition and connected with the elevation information in $\mathcal{L}$. The Retinex theory, providing the motivation for this constraint, shows that the SC is determined by the EI and has no relationship with the detailed object materials. Based on the assumption that the objects with the same LiDAR elevation always have similar EIs, we assume that the shading of one sample in $\hat{\mathcal{S}}$ can be linearly combined by its local neighbors. Similarly, $\mathcal{M}^s$ is considered for optimization since shadow also influences the EI. This constraint can be formulated as follows:

$$\begin{aligned}\mathcal{E}_3(\hat{\mathcal{S}}) = &\|\hat{\mathcal{S}} - \mathcal{W}^{L_1} \cdot (\hat{\mathcal{S}} \odot \mathcal{M}^s)\|_F^2 \\ &+ \|\hat{\mathcal{S}} - \mathcal{W}^{L_2} \cdot (\hat{\mathcal{S}} \odot \overline{\mathcal{M}^s})\|_F^2 \end{aligned} \tag{15}$$

where $\mathcal{W}^{L_1}$ and $\mathcal{W}^{L_2}$ are defined as follows:

① if $\mathcal{M}^s(\hat{\mathcal{L}}^i) = 1$ and $\mathcal{M}^s(\hat{\mathcal{L}}^i(j)) = 1$

$$\mathcal{W}^{L_1}_{i\Rightarrow j} = \begin{cases} 1 & \|\nabla\hat{\mathcal{L}}_{i\Rightarrow j}\| \le \epsilon \\ \exp\left(-\dfrac{\|\nabla\hat{\mathcal{L}}_{i\Rightarrow j}\|^2}{\delta^2}\right) & \|\nabla\hat{\mathcal{L}}_{i\Rightarrow j}\| > \epsilon \end{cases} \qquad \mathcal{W}^{L_2}_{i\Rightarrow j} = 0$$

② if $\mathcal{M}^s(\hat{\mathcal{L}}^i) + \mathcal{M}^s(\hat{\mathcal{L}}^i(j)) = 1$

$$\mathcal{W}^{L_1}_{i\Rightarrow j} = 0, \quad \mathcal{W}^{L_2}_{i\Rightarrow j} = 0$$

③ if $\mathcal{M}^s(\hat{\mathcal{L}}^i) = 0$ and $\mathcal{M}^s(\hat{\mathcal{L}}^i(j)) = 0$

$$\mathcal{W}^{L_2}_{i\Rightarrow j} = \begin{cases} 1, & \|\nabla\hat{\mathcal{L}}_{i\Rightarrow j}\| \le \epsilon \\ \exp\left(-\dfrac{\|\nabla\hat{\mathcal{L}}_{i\Rightarrow j}\|^2}{\delta^2}\right), & \|\nabla\hat{\mathcal{L}}_{i\Rightarrow j}\| > \epsilon \end{cases} \qquad \mathcal{W}^{L_1}_{i\Rightarrow j} = 0. \tag{16}$$

E. Solution of the Problem

By combining (5), (14), and (15), a novel energy function is obtained as follows:

$$\mathcal{E}(\hat{\mathcal{R}}, \hat{\mathcal{S}}) = \mathcal{E}_1(\hat{\mathcal{R}}, \hat{\mathcal{S}}) + \alpha\mathcal{E}_2(\hat{\mathcal{R}}) + \beta\mathcal{E}_3(\hat{\mathcal{S}}) \tag{17}$$

where α and β are balance coefficients. We set $\mathcal{W}^A = (\mathcal{W}^{\text{SR}} + \mathcal{W}^L)$ and $\mathcal{W}^B = (\mathcal{W}^{\text{NR}} + \mathcal{W}^L)$. Then, $\mathcal{E}_2(\hat{\mathcal{R}})$ and $\mathcal{E}_3(\hat{\mathcal{S}})$ are rewritten as follows:

$$\begin{aligned}\mathcal{E}_2(\hat{\mathcal{R}}) &= \|\hat{\mathcal{R}} \odot \mathcal{M}^s - \mathcal{W}^A\hat{\mathcal{R}}\|_F^2 + \|\hat{\mathcal{R}} \odot \overline{\mathcal{M}^s} - \mathcal{W}^B\hat{\mathcal{R}}\|_F^2 \\ \mathcal{E}_3(\hat{\mathcal{S}}) &= \|\hat{\mathcal{S}} \odot \mathcal{M}^s - \mathcal{W}^{L_1}\hat{\mathcal{S}}\|_F^2 + \|\hat{\mathcal{S}} \odot \overline{\mathcal{M}^s} - \mathcal{W}^{L_2}\hat{\mathcal{S}}\|_F^2. \end{aligned} \tag{18}$$

After that, we get the partial derivatives of $\mathcal{E}(\hat{\mathcal{R}}, \hat{\mathcal{S}})$ as follows:

$$\begin{aligned}\frac{\partial\mathcal{E}}{\partial\hat{\mathcal{R}}} = 2\big\{&\hat{\mathcal{R}} + \hat{\mathcal{S}} - \hat{\mathcal{I}} \\ &+ \alpha\big[\hat{\mathcal{R}} \odot \mathcal{M}^s \odot \mathcal{M}^s + \hat{\mathcal{R}} \odot \overline{\mathcal{M}^s} \odot \overline{\mathcal{M}^s} \\ &- \mathcal{M}^s \odot (\mathcal{W}^A\hat{\mathcal{R}}) - \overline{\mathcal{M}^s} \odot (\mathcal{W}^B\hat{\mathcal{R}}) \\ &- (\mathcal{W}^A)^T(\mathcal{M}^s \odot \hat{\mathcal{R}}) - (\mathcal{W}^B)^T(\overline{\mathcal{M}^s} \odot \hat{\mathcal{R}}) \\ &+ (\mathcal{W}^A)^T\mathcal{W}^A\hat{\mathcal{R}} + (\mathcal{W}^B)^T\mathcal{W}^B\hat{\mathcal{R}}\big]\big\} \\ \frac{\partial\mathcal{E}}{\partial\hat{\mathcal{S}}} = 2\big\{&\hat{\mathcal{R}} + \hat{\mathcal{S}} - \hat{\mathcal{I}} \\ &+ \beta\big[\hat{\mathcal{S}} \odot \mathcal{M}^s \odot \mathcal{M}^s + \hat{\mathcal{S}} \odot \overline{\mathcal{M}^s} \odot \overline{\mathcal{M}^s} \\ &- \mathcal{M}^s \odot (\mathcal{W}^{L_1}\hat{\mathcal{S}}) - \overline{\mathcal{M}^s} \odot (\mathcal{W}^{L_2}\hat{\mathcal{S}}) \\ &- (\mathcal{W}^{L_1})^T(\mathcal{M}^s \odot \hat{\mathcal{S}}) - (\mathcal{W}^{L_2})^T(\overline{\mathcal{M}^s} \odot \hat{\mathcal{S}}) \\ &+ (\mathcal{W}^{L_1})^T\mathcal{W}^{L_1}\hat{\mathcal{S}} + (\mathcal{W}^{L_2})^T\mathcal{W}^{L_2}\hat{\mathcal{S}}\big]\big\}. \end{aligned} \tag{19}$$

We refer interested readers to the appendix for (19) in detail. For obtaining the optimal solutions, an alternative updating strategy is chosen as follows:

$$\begin{aligned}\hat{\mathcal{R}}(t+1) &= \hat{\mathcal{R}}(t) - \tau\frac{\partial\mathcal{E}(\hat{\mathcal{R}}(t), \hat{\mathcal{S}}(t))}{\partial\hat{\mathcal{R}}(t)} \\ \hat{\mathcal{S}}(t+1) &= \hat{\mathcal{S}}(t) - \tau\frac{\partial\mathcal{E}(\hat{\mathcal{R}}(t+1), \hat{\mathcal{S}}(t))}{\partial\hat{\mathcal{S}}(t)} \end{aligned} \tag{20}$$

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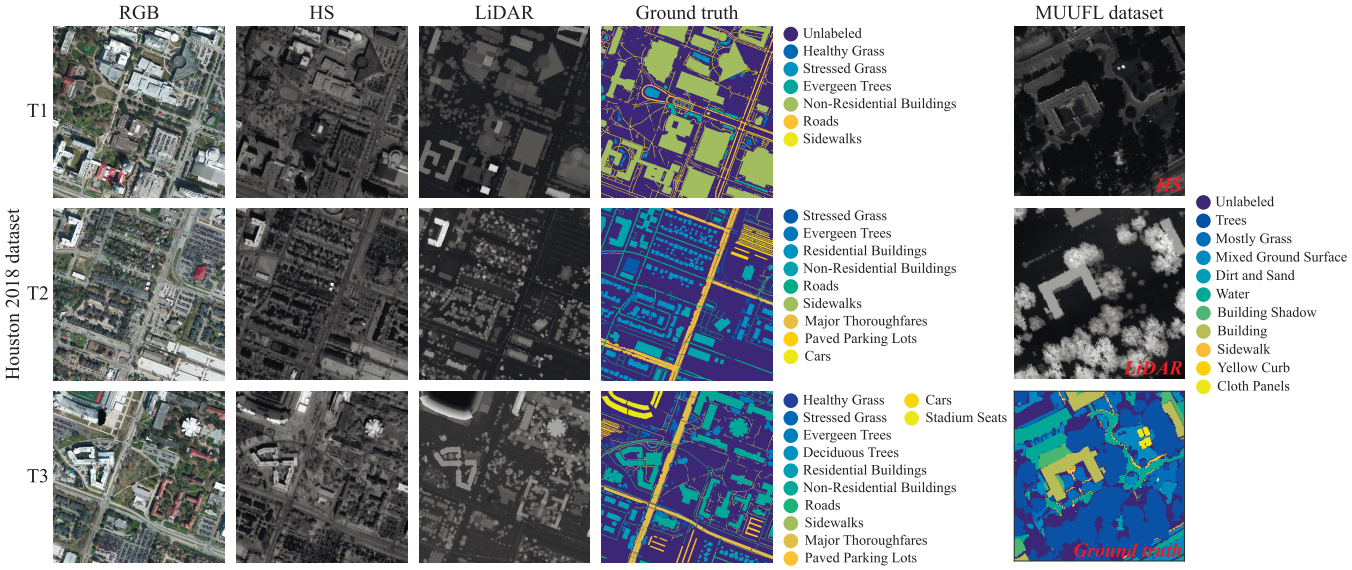


Fig. 2. Visualizations of the Houston 2018 dataset (i.e., T1, T2 and T3 subscenes) and the MUUFL dataset.

where t denotes the current iteration number. For better convergence performance, the CIS is presented for smMIID. Specifically, the initial $\hat{\mathcal{R}}$ and $\hat{\mathcal{S}}$ are set as $\hat{\mathcal{I}}$ and $\hat{\mathcal{L}}$, respectively. Notably, $\hat{\mathcal{S}}$ is initially set as $\hat{\mathcal{L}}$ rather than $0_{n\times d}$ since the elevation information in $\hat{\mathcal{L}}$ is highly correlated with the EI, providing convenience for component convergence. The energy function can converge to $\mathcal{E}(\hat{\mathcal{R}}^\star, \hat{\mathcal{S}}^\star)$.

F. Component Fusion

Existing HIID techniques tend to remove the SC and adopt the RC for the following analyzing processes [21]. However, the SC with geometric information should also be considered when accurately classifying objects. In smMIID, a simple but effective component fusion strategy is chosen to prove the SC potential. Given the optimal components $\hat{\mathcal{R}}^\star$ and $\hat{\mathcal{S}}^\star$, the fusion strategy is formulated as follows:

$$\hat{\mathcal{K}}^\star = \hat{\mathcal{R}}^\star \copyright \hat{\mathcal{S}}^\star \tag{21}$$

where © denotes the concatenation operator along the band axis. Finally, $\hat{\mathcal{K}}^\star$, $\hat{\mathcal{R}}^\star$ and $\hat{\mathcal{S}}^\star$ are transformed from the logarithmic domain to the real domain by the exponential operator as $\mathcal{K}^\star = \exp(\hat{\mathcal{K}}^\star)$, $\mathcal{R}^\star = \exp(\hat{\mathcal{R}}^\star)$ and $\mathcal{S}^\star = \exp(\hat{\mathcal{S}}^\star)$. Considering the detailed motivation in this article, other relatively complex component fusion strategies are not considered.

G. Time Complexity and Robustness

Considering the detailed calculating and optimizing procedures, $\mathcal{O} = \max\{\text{nmd}, \text{nt}_{\text{all}}\}$ is the time complexity of smMIID, where t_{all} is the total iteration number. Specifically, the running time is mainly dominated by the HS band number d when $t_{\text{all}} \leq \text{md}$. When $t_{\text{all}} > \text{md}$, t_{all} becomes the decisive factor with respect to the used time.

Shadow detection and removal are challenging tasks, reducing the credibility of the generated shadow mask. Thus, smMIID regards $\mathcal{M}^s$ as auxiliary information rather than principal information for the whole decomposition procedure. When there are deviations in $\mathcal{M}^s$, satisfying results could also be generated by smMIID, meaning that this technique is robust to shadow mask generation. For example, when all values in $\mathcal{M}^s$ are set as 1 by mistake, the optimization is simplified by removing all $\overline{\mathcal{M}^s}$ related elements. However, the fundamental idea of smMIID is still maintained in terms of the crossmodal information, guaranteeing the performance of this technique. Furthermore, the robustness of smMIID to shadow mask generation is analyzed experimentally in Section IV-E to verify the aforementioned conclusions.

IV. Experimental Results and Analysis

A. Data Description

1) Houston 2018 Dataset: It was acquired by the National Center for Airborne Laser Mapping (NCALM) over the Houston University on Feb. 16, 2017. It consists of one HS image covering a 380–1050 nm spectral range with 48 bands at a 1 m ground sampling distance (GSD) and one LiDAR data at a 50 cm GSD. The interval sampling strategy was used for the LiDAR data to unify their shapes. There are totally 20 categories. The RGB image over the same region was organized into several subscenes considering the high resolution. We took three subscenes (the RGB subscenes were only used for region selection) and their corresponding HS and LiDAR data, including: 1) UH_NAD83_273248_3289689 (abbreviated to T1); 2) UH_NAD83_272056_3289689 (abbreviated to T2); and 3) UH_NAD83_272652_3289689 (abbreviated to T3). Their visualizations and category numbers are provided in Fig. 2 and Table I, respectively. The category colors are not shared. The categories with less than 1000 samples were removed.

2) MUUFL Dataset: It was acquired from the University of Southern Mississippi Gulf Park Campus in 2010. The HS image, collected by the CASI-1500 sensor, has 64 bands. The selected HS and LiDAR subscene both contain 200 × 200 samples at a 1 m GSD. There are totally ten categories, which are listed in Table I. Fig. 2 provides a+ detailed visualization.

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TABLE I

CATEGORY NUMBERS OF THE HOUSTON 2018 DATASET (I.E., T1, T2 AND T3 SUBSCENES) AND THE MUUFL DATASET

Category	Houston 2018 (T1 Subscene)		Houston 2018 (T2 Subscene)		Houston 2018 (T3 Subscene)		MUUFL	
	Name	Number	Name	Number	Name	Number	Name	Number
1	Healthy Grass	1901	Stressed Grass	4694	Healthy Grass	1453	Trees	15193
2	Stressed Grass	3748	Evergreen Trees	1818	Stressed Grass	5071	Mostly Grass	2715
3	Evergreen Trees	5140	Residential Buildings	31709	Evergreen Trees	6435	Mixed Ground Surface	3408
4	Non-residential Buildings	112402	Non-residential Buildings	26037	Deciduous Trees	2955	Dirt and Sand	160
5	Roads	13589	Roads	17597	Residential Buildings	8063	Road	4343
6	Sidewalks	16929	Sidewalks	1560	Non-Residential Buildings	48264	Building Shadow	1211
7	/	/	Major Thoroughfares	20504	Roads	7039	Building	2828
8	/	/	Paved Parking Lots	3490	Sidewalks	12497	Sidewalk	366
9	/	/	Cars	2625	Major Thoroughfares	17901	Yellow Curb	169
10	/	/	/	/	Paved Parking Lots	4156	Cloth Panels	269
11	/	/	/	/	Cars	1101	/	/
12	/	/	/	/	Stadium Seats	6824	/	/

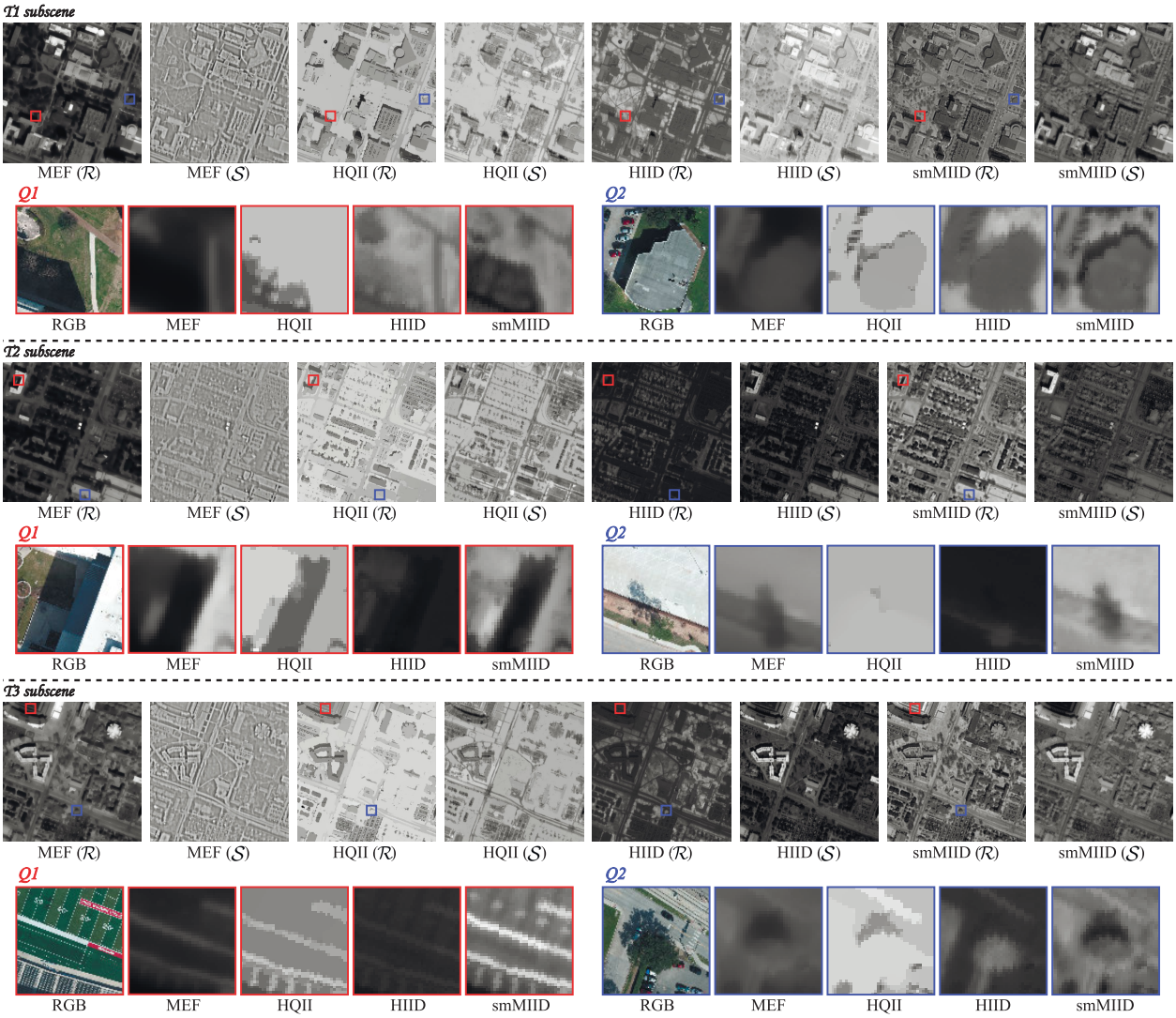


Fig. 3. Visualization results by IID-wise techniques on the Houston 2018 dataset with three subscenes (i.e., T1, T2, and T3). Two typical regions, including Q1 (red squares) and Q2 (blue squares), in the $\mathcal{R}$s are enlarged for a clear comparison. Their corresponding RGB regions are provided for reference.

B. Experimental Settings

All experiments were carried out on a PC (16 GB RAM) with Intel Core i7-8750 CPU. The codes were implemented using MATLAB R2016a. The initial HS and LiDAR datasets were normalized into $[0, 1]$. The spatial neighbor number m was selected within $[3, 5, 7, 9, 11, 13]$ (the default m is 5). The balance coefficients α and β were both selected within $[0.1, 0.3, 0.5, 0.7, 0.9]$ (the default α and β are both 0.5). The learning rate LR was selected within $[10^{-1}, 10^{-2}, 10^{-3}]$ (the default LR is 10^{-2}). Inspired by other related works [19], [21], the visualization and classification performance of smMIID was compared with fifteen state-of-the-art techniques, including the morphological attribute profiles (MAPs) [3], discriminative graph-based fusion (DGF) [1],

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accelerated adaptive feature balance technique (A^2FBT) [30], tensor singular spectral analysis (TSSA) [6], multiscale spectral-spatial feature extraction (MSF) [7], multiexposure fusion network (MEF) [17], HQII [18], HIID [21], three HS and LiDAR data fusion baselines [11] (i.e., early fusion, middle fusion and late fusion), EndNet [11], IPCNN [12], feature fusion and extraction framework (FusAtNet) [31] and multibranch feature fusion network with self- and cross-guided attention (MB2FscgaNet) [4]. Their detailed structures and hyper-parameters were fine-tuned based on their corresponding articles and our settings. Notably, MEF, HQII, and HIID are IID-wise techniques. Furthermore, the normalized HS and LiDAR datasets were also classified for a fair comparison (denoted as ORIGIN). Three effective classifiers, including the support vector machine (SVM), discriminant analysis (DA), and random forest (RF), were used to evaluate the performance quantitatively. To remove the effect of training set selection, totally 1% and 5% samples in each category were randomly selected for training, respectively. The remaining samples were used for testing. The classification performance was evaluated by three evaluation metrics [19], [21], including the overall accuracy (OA), kappa coefficient (κ), and precision (PR). They are formatted as

$$\mathrm{OA} = \frac{n_{\mathrm{correct}}}{n_{\mathrm{all}}}, \quad \kappa = \frac{\mathrm{OA} - \mathrm{Acc}}{1 - \mathrm{Acc}}$$
$$\mathrm{PR} = \frac{1}{c}\sum_{i}^{c} \frac{n^i_{\mathrm{correct}}}{n^i_{\mathrm{correct}} + \mathrm{FP}^i} \tag{22}$$

where n_{correct} is the correctly classified sample number, n_{all} is the total sample number, $\mathrm{Acc} = \sum_i^c (n^i_{\mathrm{all}} \times \hat{n}^i_{\mathrm{all}})/(n_{\mathrm{all}} \times n_{\mathrm{all}})$, n^i_{all} and $\hat{n}^i_{\mathrm{all}}$ are the sample numbers in the ith category of the real and classified labels, n^i_{correct} and FP^i are the numbers of the correctly classified samples in the ith category and the samples wrongly classified into the ith category.

C. Experimental Results and Comparison

In this section, the experimental results were evaluated by visualization and classification on the four subscenes. Specifically, better results always have higher visual qualities with distinct edges and can be classified more accurately because of their comprehensive land-cover descriptions.

1) Visualization Performance Comparison: The visualization results by all IID-wise techniques on the Houston 2018 and MUUFL datasets are provided in Figs. 3 and 4, respectively. Typical channels in $\mathcal{R}$s and $\mathcal{S}$s are shown for a clear and fair comparison. In Fig. 3, two typical regions (i.e., Q1 and Q2) in $\mathcal{R}$s and their corresponding RGB regions are enlarged. It is worth noting that MEF and HQII are challenged when capturing some on-ground objects accurately since they do not fully consider and utilize the spectral channel correlation. For example, the shadow profile of the tree in T3 (Q2) can not be clearly distinguished by MEF and HQII. The results illustrate that the decomposition performance using one single HS band is always restricted by the detailed wavelength. HIID is capable of generating high-quality decomposition components since it considers multiple HS-based constraints on $\mathcal{R}$. Apparently, smMIID yields the best performance when recovering its $\mathcal{R}$s and achieves high consistency within categories. Furthermore, the $\mathcal{S}$s of smMIID describe the detailed illumination in the specific scenarios. Considering that smMIID introduces a shadow mask during the optimization process, its components have distinct edges between shadow and nonshadow regions. In real-world tasks, accurately detecting shadow regions is helpful for land-cover analyzing (*Building Shadow* is one category in the MUUFL dataset notably). The visualization performance of smMIID is satisfactory. The obtained results further reflect that smMIID is capable of preserving abundant spatial information in $\mathcal{S}$, which is beneficial to classification tasks. Similar conclusions could be drawn from Fig. 4.

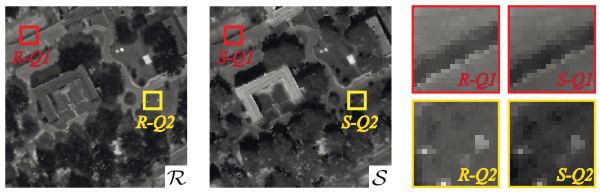


Fig. 4. Visualization results of smMIID on the MUUFL dataset. Two typical regions, including Q1 (red squares) and Q2 (blue squares), are enlarged.

2) Classification Performance Comparison: The classification results on the three Houston 2018 subscenes and the MUUFL subscene are shown in Tables II–V. Specifically, for the non-DL techniques, all three classifiers (i.e., SVM, DA, and RF) are used for evaluation. Considering that these DL techniques contain one Softmax activated fully connected layer for classification, their accuracies are directly compared with the SVM-based accuracies of other non-DL techniques. For the IID-wise techniques, their $\mathcal{R}$s, $\mathcal{S}$s, and $\mathcal{K}$s are all classified for a clear comparison. The bold values indicate the best accuracies per row. The underlined values are the best accuracies in all $\mathcal{R}$s/$\mathcal{S}$s/$\mathcal{K}$s of the IID-wise techniques. The obtained results illustrate that smMIID outperforms other techniques in terms of all evaluation metrics simultaneously. It shows that smMIID is capable of generating components with better classification performance. Most HS and LiDAR fusion techniques are capable of generating results with higher accuracies than the techniques using only one single RS modality (e.g., MAP), showing the feasibility and necessity of fusing HS and LiDAR data. Notably, smMIID outperforms the multimodal DL networks (e.g., IPCNN, FusAtNet, and MB2FscgaNet), verifying its effective learning ability. The classification performance using all classifiers is consistent, removing the effect of classifier selection. When the training sample percentage increases from 1% to 5%, most accuracies increase since more labeled samples are considered for training. Considering $\mathcal{L}$ contains one band, its classification ability is relatively poorer and keeps stable with the increase of the training sample number. Correspondingly, when fusing $\mathcal{I}_{\mathcal{HS}}$ and $\mathcal{L}$ directly, the obtained results are not satisfactory. Since smMIID considers the constraints on both $\mathcal{R}$ and $\mathcal{S}$, its performance significantly improves when fusing both components.

The classification maps by SVM with 1% training samples of all non-DL techniques are shown in Fig. 5 (the maps of DL techniques are not provided for simplification).

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TABLE II

CLASSIFICATION RESULTS, INCLUDING OA (%), κ ($\times$100) AND PR (%), ON THE T1 SUBSCENE OF THE HOUSTON 2018 DATASET

Method	1%			5%			1%			5%			1%			5%		
	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR
Early Fusion	89.62	76.68	74.81	91.17	80.05	81.31	/	/	/	/	/	/	/	/	/	/	/	/
Middle Fusion	88.91	75.14	76.47	91.57	80.89	82.01	/	/	/	/	/	/	/	/	/	/	/	/
Late Fusion	88.99	75.58	74.81	91.28	80.23	80.18	/	/	/	/	/	/	/	/	/	/	/	/
EndNet	87.94	73.01	73.84	90.86	79.61	81.96	/	/	/	/	/	/	/	/	/	/	/	/
IPCNN	89.33	75.53	74.18	91.70	80.36	84.13	/	/	/	/	/	/	/	/	/	/	/	/
FusAtNet	90.03	77.44	82.28	92.12	82.36	83.63	/	/	/	/	/	/	/	/	/	/	/	/
MB2FscgaNet	89.95	76.75	**84.62**	92.31	**82.82**	83.16	/	/	/	/	/	/	/	/	/	/	/	/
Classifier	SVM						DA						RF					
ORIGIN ($\mathcal{I}_{\mathcal{HS}}$)	86.19	66.40	74.23	88.07	71.91	76.32	83.47	60.24	60.08	83.97	61.48	61.02	84.12	62.41	68.74	85.54	65.02	73.27
ORIGIN ($\mathcal{L}$)	80.71	54.51	63.08	80.37	53.91	58.78	80.29	53.49	66.83	80.39	53.00	66.56	78.16	50.30	39.23	78.03	50.93	38.96
ORIGIN ($\mathcal{K}$)	88.45	72.95	82.44	89.74	76.49	76.09	85.80	68.29	61.32	85.98	68.93	61.31	88.68	73.66	74.27	89.18	75.06	75.09
MAP	87.53	70.47	78.13	89.54	75.60	80.26	83.57	60.67	59.63	84.15	62.19	60.89	85.13	64.56	71.70	88.48	72.81	77.59
DGF	88.13	71.89	75.43	90.39	77.75	81.55	84.13	62.54	61.10	84.20	62.34	60.81	86.28	67.24	73.22	89.94	76.46	81.26
A^2FBT	81.75	54.24	57.80	83.29	57.92	66.98	80.88	53.70	52.62	81.00	52.93	53.66	84.34	61.91	69.52	87.24	69.76	75.58
TSSA	87.45	70.53	74.27	89.39	75.49	77.96	86.34	68.83	64.74	86.64	69.67	64.38	84.29	61.73	68.32	87.22	69.49	74.75
MSF	87.19	69.51	75.05	91.00	78.99	82.28	85.78	66.69	64.45	86.11	67.36	66.26	89.32	74.55	**80.02**	91.22	79.43	82.75
MEF ($\mathcal{R}$)	85.83	66.07	74.72	87.69	70.83	75.59	83.03	61.07	56.39	84.13	62.96	59.95	85.64	65.25	73.99	87.22	69.80	75.15
MEF ($\mathcal{S}$)	81.42	51.54	64.79	83.46	57.83	68.56	78.70	41.99	55.00	78.54	42.61	52.80	77.30	39.61	46.87	80.19	47.95	57.21
MEF ($\mathcal{K}$)	87.18	69.51	75.92	88.98	74.15	78.02	86.34	68.41	63.95	86.81	69.32	67.12	85.78	66.11	72.31	89.48	75.15	80.84
HQII ($\mathcal{R}$)	86.43	68.59	71.45	89.38	75.22	78.87	82.25	59.07	60.08	82.95	59.88	62.45	84.03	60.46	67.90	87.34	69.36	76.84
HQII ($\mathcal{S}$)	85.54	65.04	73.36	89.00	74.33	77.99	82.53	58.72	60.95	83.41	60.66	63.29	83.78	60.24	67.10	85.73	65.68	72.46
HQII ($\mathcal{K}$)	89.64	76.14	78.74	92.17	82.08	84.52	85.53	67.05	64.01	86.73	69.92	67.45	88.58	73.04	78.36	90.78	78.53	**83.06**
HIID ($\mathcal{R}$)	86.61	68.40	73.73	88.99	74.33	78.12	83.46	60.28	63.31	83.06	58.66	62.63	84.25	61.52	68.88	87.47	70.33	72.64
HIID ($\mathcal{S}$)	82.85	56.57	69.67	86.65	67.82	78.03	82.53	58.22	61.61	83.49	60.19	63.16	84.97	63.56	66.70	87.18	69.42	75.02
HIID ($\mathcal{K}$)	87.19	69.88	75.28	89.79	76.31	80.67	82.46	57.80	60.49	82.75	57.90	62.23	86.28	67.78	71.42	88.89	74.09	77.68
smMIID ($\mathcal{R}$)	88.39	73.04	75.21	90.31	77.67	79.45	86.01	67.83	64.50	86.25	68.37	64.82	86.30	67.86	70.39	88.35	72.93	75.13
smMIID ($\mathcal{S}$)	88.48	73.16	76.43	90.75	78.73	82.38	86.76	70.26	65.09	86.79	70.48	64.15	86.24	67.58	73.80	88.96	74.52	78.15
smMIID ($\mathcal{K}$)	**90.72**	**78.78**	81.09	**92.43**	**82.82**	**84.88**	**89.08**	**75.78**	**72.34**	**89.34**	**76.36**	**72.83**	**89.51**	**76.38**	77.84	**91.23**	**79.96**	82.04

TABLE III

CLASSIFICATION RESULTS, INCLUDING OA (%), κ ($\times$100) AND PR (%), ON THE T2 SUBSCENE OF THE HOUSTON 2018 DATASET

Method	1%			5%			1%			5%			1%			5%		
	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR
Early Fusion	76.97	71.19	69.84	82.70	78.28	77.44	/	/	/	/	/	/	/	/	/	/	/	/
Middle Fusion	73.42	66.60	69.06	83.12	78.75	79.76	/	/	/	/	/	/	/	/	/	/	/	/
Late Fusion	74.25	67.79	69.03	80.88	75.98	76.51	/	/	/	/	/	/	/	/	/	/	/	/
EndNet	66.39	57.98	58.16	79.45	74.17	73.88	/	/	/	/	/	/	/	/	/	/	/	/
IPCNN	68.57	60.20	64.68	83.18	78.99	80.01	/	/	/	/	/	/	/	/	/	/	/	/
FusAtNet	80.57	75.49	78.17	84.30	80.30	80.86	/	/	/	/	/	/	/	/	/	/	/	/
MB2FscgaNet	78.99	73.62	77.39	84.03	79.96	80.71	/	/	/	/	/	/	/	/	/	/	/	/
Classifier	SVM						DA						RF					
ORIGIN ($\mathcal{I}_{\mathcal{HS}}$)	74.10	67.09	68.22	79.51	74.04	79.71	63.92	54.23	59.03	67.68	59.10	64.34	67.11	58.19	58.89	73.69	66.77	68.72
ORIGIN ($\mathcal{L}$)	50.49	35.38	56.86	54.39	40.92	54.48	45.04	28.58	48.11	44.46	27.88	47.00	45.46	31.68	28.93	46.38	32.57	28.55
ORIGIN ($\mathcal{K}$)	74.56	67.75	74.66	81.15	76.20	81.80	67.71	59.33	59.64	70.62	63.09	65.93	72.13	64.76	62.97	79.08	73.63	74.72
MAP	77.51	71.52	74.51	81.80	77.00	80.98	66.78	58.18	57.51	69.54	61.50	64.53	70.12	62.17	64.76	78.21	72.42	**76.86**
DGF	79.01	73.39	77.53	83.86	79.61	83.70	68.10	59.66	61.19	70.14	62.27	64.71	70.89	63.10	65.30	78.87	73.32	75.34
A^2FBT	76.64	70.47	76.74	78.74	73.16	78.07	67.14	58.93	62.48	68.64	60.81	64.04	66.60	57.56	59.65	73.84	66.89	68.79
TSSA	77.15	71.12	77.66	81.07	76.10	80.37	**78.04**	72.57	**72.03**	**78.35**	**73.00**	70.06	71.86	64.36	66.05	78.03	72.22	75.84
MSF	78.74	73.10	76.37	81.22	76.21	81.26	75.68	69.23	70.61	76.20	69.92	71.51	73.23	**66.54**	66.57	79.15	**73.85**	71.75
MEF ($\mathcal{R}$)	71.24	63.66	69.50	77.19	71.19	75.53	69.87	62.42	61.32	71.40	64.29	61.82	71.95	64.54	67.69	77.18	71.29	69.71
MEF ($\mathcal{S}$)	54.26	41.02	56.20	61.23	50.50	61.33	46.45	32.49	38.77	47.23	32.56	37.93	45.94	30.97	41.02	47.87	33.55	41.12
MEF ($\mathcal{K}$)	72.95	65.82	72.16	78.67	73.07	76.68	71.08	63.62	67.08	73.67	66.87	69.62	72.32	65.06	65.69	77.36	71.49	73.19
HQII ($\mathcal{R}$)	77.22	71.25	68.19	80.88	75.89	80.68	65.47	57.11	53.75	69.11	61.52	58.80	69.38	61.26	62.23	74.00	67.15	69.40
HQII ($\mathcal{S}$)	72.32	65.05	75.92	76.41	70.25	77.56	65.55	57.05	58.79	68.37	60.35	64.45	68.36	60.04	63.45	74.22	67.51	71.37
HQII ($\mathcal{K}$)	80.78	75.70	80.62	84.03	79.89	83.10	75.07	68.96	65.20	77.38	71.80	69.13	73.18	66.21	67.98	77.16	71.21	74.97
HIID ($\mathcal{R}$)	76.15	69.82	78.10	80.12	74.92	79.84	64.03	55.10	53.72	65.03	56.06	57.85	67.69	59.18	60.53	74.83	68.28	68.87
HIID ($\mathcal{S}$)	72.38	64.98	72.95	76.26	69.96	76.44	65.54	56.44	60.65	66.40	57.62	61.78	71.44	63.96	67.73	75.28	68.84	71.82
HIID ($\mathcal{K}$)	77.81	71.89	78.75	84.41	80.33	**83.99**	69.11	61.40	63.69	72.99	66.26	66.52	72.99	65.97	67.72	77.99	72.26	75.63
smMIID ($\mathcal{R}$)	77.33	71.34	74.03	81.68	76.87	82.06	72.78	65.82	66.60	74.59	68.18	68.75	70.29	62.46	62.73	75.41	68.98	69.73
smMIID ($\mathcal{S}$)	78.94	73.43	79.85	81.45	76.61	82.88	69.80	62.15	65.44	71.64	64.41	69.12	69.75	61.75	**68.20**	75.80	69.44	73.06
smMIID ($\mathcal{K}$)	**82.13**	**77.47**	**82.01**	**84.51**	**80.47**	83.75	75.92	**69.91**	70.18	78.22	72.80	**72.49**	**73.42**	**66.54**	67.92	**79.16**	73.73	76.59

For the IID-wise techniques, their $\mathcal{K}$ maps are shown. The unlabeled background samples with noise are not considered for evaluation. They are also not classified in other related works [32], [33]. The results clearly illustrate that smMIID is capable of generating a more accurate classification map with fewer wrongly classified samples than other techniques. Furthermore, smMIID outperforms other techniques with regard to intraclass consistency. For example, the *Non-Residential*

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TABLE IV

CLASSIFICATION RESULTS, INCLUDING OA (%), κ ($\times$100) AND PR (%), ON THE T3 SUBSCENE OF THE HOUSTON 2018 DATASET

Method	1%			5%			1%			5%			1%			5%		
	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR
Early Fusion	76.37	70.02	69.37	83.42	79.11	75.97	/	/	/	/	/	/	/	/	/	/	/	/
Middle Fusion	77.31	71.20	70.16	83.62	79.30	77.78	/	/	/	/	/	/	/	/	/	/	/	/
Late Fusion	76.29	70.09	67.87	83.23	78.85	77.67	/	/	/	/	/	/	/	/	/	/	/	/
EndNet	75.42	68.83	63.64	84.74	80.77	79.02	/	/	/	/	/	/	/	/	/	/	/	/
IPCNN	74.91	68.39	67.15	83.87	79.34	80.62	/	/	/	/	/	/	/	/	/	/	/	/
FusAtNet	78.76	73.45	75.71	84.83	80.93	81.16	/	/	/	/	/	/	/	/	/	/	/	/
MB2FscgaNet	79.83	74.76	76.92	85.56	81.85	81.82	/	/	/	/	/	/	/	/	/	/	/	/
Classifier	SVM						DA						RF					
ORIGIN ($\mathcal{I}_{\mathcal{HS}}$)	73.91	66.50	67.37	80.52	75.11	76.26	69.02	60.75	57.86	71.68	64.17	61.05	67.80	59.00	55.73	75.22	68.48	66.41
ORIGIN ($\mathcal{L}$)	55.89	38.17	58.69	56.04	39.13	54.51	51.00	32.04	66.36	50.80	31.48	49.32	48.29	34.78	28.44	48.19	34.56	27.50
ORIGIN ($\mathcal{K}$)	76.34	69.67	71.04	83.73	79.38	79.46	72.09	65.05	60.16	73.87	67.28	62.68	73.84	66.71	63.74	**80.18**	**74.87**	72.44
MAP	75.32	68.46	69.07	82.33	77.55	78.34	74.32	67.67	62.20	75.48	69.30	67.10	69.98	61.69	59.53	73.15	65.71	63.35
DGF	78.76	73.01	71.59	82.80	78.15	78.62	76.60	70.76	68.05	78.52	73.13	69.08	71.39	63.48	60.16	78.74	72.93	72.16
A^2FBT	76.94	70.68	71.94	79.88	74.49	74.90	73.42	66.64	62.08	74.09	67.44	63.34	68.35	59.26	56.69	75.35	68.57	66.31
TSSA	76.52	70.12	74.27	81.85	76.98	77.42	77.12	71.48	67.10	80.04	75.05	**72.32**	71.66	63.68	62.04	77.73	71.59	70.80
MSF	79.79	74.13	75.84	85.30	81.28	82.09	77.26	71.51	68.03	80.11	75.06	71.74	74.61	68.05	66.45	78.72	73.20	72.95
MEF ($\mathcal{R}$)	74.27	66.73	68.41	77.68	71.53	72.19	74.08	67.55	63.43	75.50	69.02	66.54	69.88	61.50	58.15	74.57	67.69	69.30
MEF ($\mathcal{S}$)	53.54	35.71	49.88	60.07	46.49	56.96	48.04	28.70	36.09	48.80	29.56	37.90	45.90	25.69	29.87	47.52	29.36	33.01
MEF ($\mathcal{K}$)	74.24	66.49	71.56	79.77	74.15	75.20	76.34	70.27	67.75	77.27	71.49	69.53	73.24	65.58	**68.11**	77.86	71.75	**73.80**
HQII ($\mathcal{R}$)	75.27	68.46	67.67	81.75	76.82	78.73	71.64	64.32	59.35	72.47	65.55	61.64	70.23	62.20	57.18	74.98	68.41	65.32
HQII ($\mathcal{S}$)	73.20	65.13	71.63	79.88	74.31	78.21	67.65	58.84	58.05	70.02	61.91	62.46	71.86	64.37	62.12	73.40	66.15	63.79
HQII ($\mathcal{K}$)	80.16	74.56	76.88	85.62	81.78	**83.47**	77.42	71.72	68.34	79.07	73.79	70.10	74.93	68.22	66.68	77.24	71.21	69.66
HIID ($\mathcal{R}$)	73.41	65.80	65.20	80.59	75.26	75.83	71.23	63.92	59.95	73.04	66.02	63.39	69.77	61.57	58.07	75.17	68.45	65.30
HIID ($\mathcal{S}$)	69.89	60.78	63.72	78.99	73.06	77.14	69.70	61.80	58.85	71.32	63.50	62.34	72.34	64.76	62.69	76.18	69.83	67.59
HIID ($\mathcal{K}$)	76.76	70.30	72.12	84.16	79.87	81.35	73.82	67.16	62.37	76.20	70.08	66.38	76.11	69.70	68.04	79.05	73.46	72.82
smMIID ($\mathcal{R}$)	75.85	69.01	69.27	82.27	77.46	76.92	74.12	67.31	62.50	75.28	68.86	63.53	70.90	63.19	61.23	75.23	68.53	66.80
smMIID ($\mathcal{S}$)	78.09	72.21	72.53	83.59	79.17	80.56	74.79	68.37	66.90	75.55	69.37	66.94	72.39	64.80	63.95	77.25	71.15	69.44
smMIID ($\mathcal{K}$)	**80.24**	**74.92**	**76.95**	**85.95**	**82.15**	81.92	**77.54**	**71.77**	**68.56**	**80.17**	**75.15**	72.31	**76.36**	**69.96**	67.41	79.16	73.47	73.42

TABLE V

CLASSIFICATION RESULTS, INCLUDING OA (%), κ ($\times$100), AND PR (%), ON THE MUUFL DATASET

Method	1%			5%			1%			5%			1%			5%		
	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR	OA	κ	PR
Early Fusion	87.22	81.69	77.92	90.50	86.53	82.62	/	/	/	/	/	/	/	/	/	/	/	/
Middle Fusion	86.96	81.46	62.01	87.96	82.83	76.46	/	/	/	/	/	/	/	/	/	/	/	/
Late Fusion	88.03	82.93	77.40	90.54	86.45	83.25	/	/	/	/	/	/	/	/	/	/	/	/
EndNet	87.80	82.75	75.53	89.66	85.33	79.67	/	/	/	/	/	/	/	/	/	/	/	/
IPCNN	88.43	83.49	70.70	91.18	87.39	76.12	/	/	/	/	/	/	/	/	/	/	/	/
FusAtNet	88.65	83.58	73.63	91.83	88.32	74.11	/	/	/	/	/	/	/	/	/	/	/	/
MB2FscgaNet	88.98	84.11	70.66	92.41	89.15	77.01	/	/	/	/	/	/	/	/	/	/	/	/
Classifier	SVM						DA						RF					
ORIGIN ($\mathcal{I}_{\mathcal{HS}}$)	84.81	78.13	78.97	84.96	78.01	81.95	83.77	77.12	72.34	86.67	81.19	76.29	79.36	72.76	74.18	84.31	79.19	78.60
ORIGIN ($\mathcal{L}$)	55.44	32.70	51.52	60.22	40.47	54.93	53.83	32.02	53.51	54.64	32.75	53.65	47.97	30.92	27.77	47.40	30.53	28.61
ORIGIN ($\mathcal{K}$)	85.86	79.40	82.87	87.86	82.32	77.54	84.12	77.60	74.14	88.08	83.27	77.04	84.45	79.36	76.92	87.36	83.22	83.58
MAP	86.64	80.95	84.42	87.35	81.97	76.74	84.78	78.44	69.79	86.78	81.34	76.49	82.60	76.93	69.77	88.08	84.15	80.12
DGF	88.01	82.81	83.69	91.28	87.53	86.54	85.33	79.26	72.42	87.09	81.68	77.94	83.31	77.84	73.42	87.57	83.55	79.84
A^2FBT	83.88	76.64	77.46	90.13	85.87	88.76	81.43	73.71	65.64	88.29	83.40	78.90	82.94	75.31	76.99	87.82	82.51	82.50
TSSA	87.97	82.68	84.85	90.73	86.75	**89.77**	86.59	80.18	76.59	90.14	85.52	**84.95**	83.27	75.66	64.36	87.35	82.11	71.88
MSF	86.31	80.33	83.55	92.03	88.59	86.42	86.43	80.72	73.17	90.42	86.50	74.59	83.79	76.59	76.78	88.89	84.04	78.29
MEF ($\mathcal{R}$)	83.66	76.51	80.10	88.67	83.72	81.26	84.31	77.39	66.87	86.60	80.77	77.19	80.54	71.29	67.02	86.42	80.29	73.71
MEF ($\mathcal{S}$)	64.00	43.68	57.72	70.89	56.50	63.60	61.79	44.76	41.94	65.31	48.98	49.08	57.90	35.88	52.58	61.31	40.74	50.15
MEF ($\mathcal{K}$)	84.51	77.38	74.42	89.84	85.45	80.11	86.48	80.79	75.36	90.50	86.54	80.25	81.39	73.27	65.01	86.56	80.58	72.32
HQII ($\mathcal{R}$)	85.05	78.42	83.26	87.48	82.11	70.98	74.79	64.86	56.64	82.67	75.65	71.72	80.59	74.23	74.02	87.22	83.02	84.11
HQII ($\mathcal{S}$)	85.35	78.95	76.68	87.74	82.50	81.18	79.47	71.10	59.10	83.88	77.02	67.72	73.63	64.29	70.14	79.09	71.79	74.66
HQII ($\mathcal{K}$)	84.68	77.72	82.47	88.06	82.94	81.44	81.96	74.76	67.97	88.65	83.91	80.02	84.07	78.65	75.73	88.32	84.50	82.26
HIID ($\mathcal{R}$)	85.19	78.73	76.08	88.05	82.83	83.82	77.98	69.62	61.95	83.35	76.69	70.01	80.29	73.70	70.45	83.53	78.19	77.75
HIID ($\mathcal{S}$)	84.05	77.03	82.25	88.08	82.92	82.76	80.17	72.22	63.24	84.09	77.61	70.99	78.69	71.59	69.19	82.96	77.45	77.26
HIID ($\mathcal{K}$)	87.89	82.62	**88.52**	91.80	88.32	86.12	80.70	72.95	63.39	85.90	80.17	73.68	81.52	75.30	74.61	84.28	79.10	81.20
smMIID ($\mathcal{R}$)	86.17	80.35	78.44	89.43	84.87	85.72	83.31	76.66	74.66	86.66	81.28	74.71	81.80	75.47	77.89	84.30	79.14	80.63
smMIID ($\mathcal{S}$)	88.87	83.94	88.22	91.01	87.13	84.54	84.92	78.99	71.54	88.24	83.59	76.84	80.42	73.71	77.40	86.00	81.42	78.70
smMIID ($\mathcal{K}$)	**89.05**	**84.13**	79.75	**92.43**	**89.18**	89.16	**86.60**	**81.10**	**81.07**	**90.71**	**86.98**	83.65	**84.69**	**79.66**	**79.33**	**88.94**	**85.29**	**84.97**

Building samples in the T3 subscene can be classified better by smMIID than other comparison techniques.

Additional experiments were conducted on the whole Houston 2018 dataset. The train-test data split strategy in [31] is selected. Although this strategy was specifically presented for the Houston 2013 dataset (acquired in 2013), it is still quotable when using the Houston 2018 dataset. Notably, these two datasets were captured in different years and have diverse

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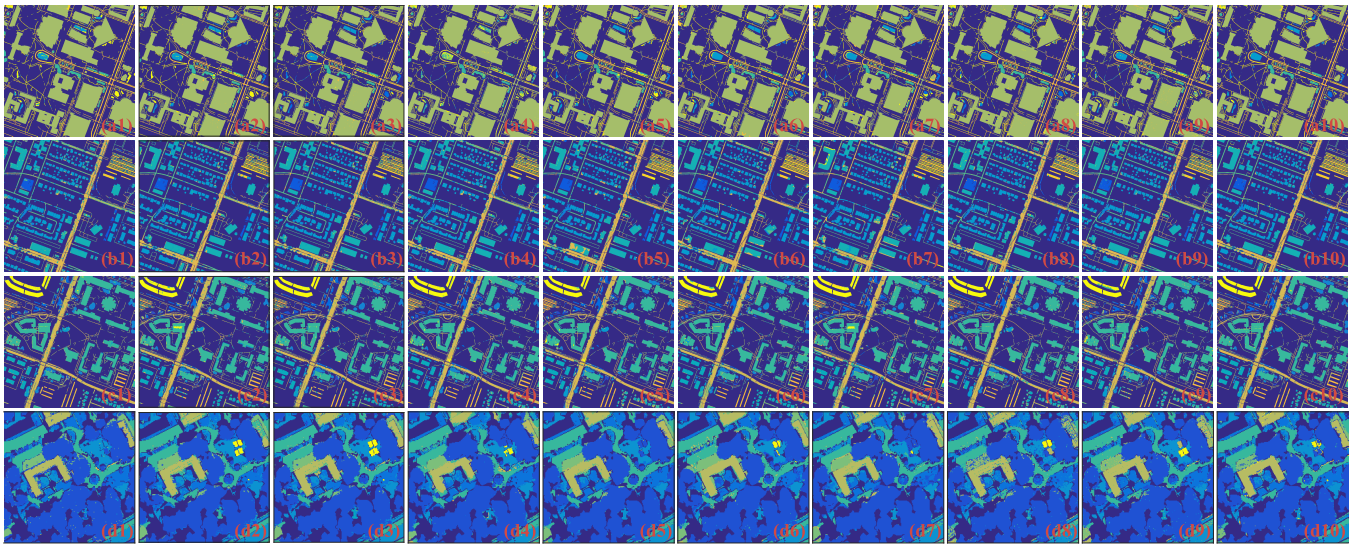


Fig. 5. Classification maps by SVM with 1% training samples of all non-DL techniques, including: (a) T1 subscene, (b) T2 subscene, (c) T3 subscene, and (d) MUUFL dataset; (1) ORIGIN, (2) MAP, (3) DGF, (4) A^{2}FBT, (5) TSSA, (6) MSF, (7) MEF, (8) HQII, (9) HIID, and (10) smMIID.

TABLE VI

CLASSIFICATION RESULTS, INCLUDING OA (%), κ ($\times$100), AND PR (%), ON THE WHOLE HOUSTON 2018 DATASET

Accuracy	EF	MF	LF	EndNet	IPCNN	FusAtNet	MB2FNet	MAP
OA	75.27	74.05	74.96	81.34	82.36	83.43	83.42	80.23
κ	70.24	68.28	69.79	77.69	78.80	80.14	80.11	76.38
PR	65.04	66.33	66.87	71.84	72.65	**74.08**	73.21	64.29
Accuracy	DCF	A^{2}FBT	TSSA	MSF	MEF	HQII	HIID	smMIID
OA	80.98	72.03	73.65	77.52	82.16	82.65	81.83	**83.47**
κ	77.27	65.68	68.15	72.90	78.70	79.14	78.05	**80.17**
PR	69.53	62.37	61.81	69.20	72.31	71.92	74.00	73.06

TABLE VII

CLASSIFICATION RESULTS, INCLUDING OA (%), κ ($\times$100), PR (%) AND CALCULATION TIME ($\times 10^{-3}$ s) PER SAMPLE, WITH DIFFERENT mS ON THE T1 SUBSCENE IN THE HOUSTON 2018 DATASET

Category	3	5	7	9	11	13
1	**74.91**	64.49	72.99	62.89	64.27	59.22
2	80.59	80.89	83.94	83.02	**85.61**	84.31
3	85.79	89.94	89.62	**90.53**	89.50	90.31
4	97.20	98.30	**98.34**	98.06	98.29	98.17
5	57.96	67.51	66.40	**67.62**	66.46	67.48
6	59.31	**64.39**	63.48	61.62	63.02	60.93
OA	88.49	**90.72**	**90.72**	90.30	90.57	90.27
κ	73.52	78.78	**78.86**	78.05	78.57	77.92
PR	76.88	**81.09**	78.41	76.65	78.00	76.95
Time	6.744	13.45	21.98	32.54	55.48	**71.42**

GSDs, data shapes, and HS band numbers. Thus, their overlapping HS and LiDAR regions with 207 $\times$ 954 samples were aligned and chosen manually. The interval sampling strategy was used to unify the GSDs. Considering some categories in the Houston 2013 dataset have disappeared, the shared categories in both the training and testing sets were retained. There are 16 categories in the overlapping dataset. Their training|testing sample numbers are 5|57 (*Healthy Grass*), 51|293 (*Stressed Grass*), 13|56 (*Evergreen Trees*), 6|9 (*Deciduous Trees*), 3|7 (*Water*), 36|148 (*Residential Buildings*), 184|645 (*Non-Residential Buildings*), 18|133 (*Roads*), 12|55 (*Sidewalks*), 3|8 (*Crosswalks*), 6|35 (*Major Thoroughfares*), 8|7 (*Highways*), 34|211 (*Railways*), 9|73 (*Paved Parking Lots*), 9|98 (*Cars*), and 6|62 (*Trains*), respectively. The classification results of all methods are shown in Table VI. The three baselines and MB2FscgaNet are abbreviated to EF, MF, LF, and MB2FNet, respectively. Some methods, including FusAtNet, MB2Fnet, and smMIID, outperform other methods in terms of OA. Although some methods (i.e., FusAtNet, MB2Fnet, and HIID) have higher PRs, smMIID performs the best comprehensively in terms of the three evaluation metrics, showing that smMIID is capable of generating competitive fusion results.

3) Influence of the Spatial Neighbor Number m: The experiments were conducted with 1% training samples on the T1 subscene of the Houston 2018 dataset to analyze the influences of the parameters in smMIID. The spatial neighbor number m varied within [3, 5, 7, 9, 11, 13] for optimal value selection. The classification results and calculation time per sample when generating the correlation matrices are recorded in Table VII. Apparently, more spatial information is considered for training when m is large. However, large ms could increase the combination challenge of neighboring samples and calculation time per epoch. Specifically, the biggest OA is obtained when m is set as 5 or 7. Furthermore, when $m = 5$ and $m = 7$, κ and PR reach their maximums, respectively. When m increases from 5 to 7, the calculation time per sample increases 8.53×10^{-3} s, making the whole process time-consuming. The optimal value selection should consider the effectiveness and efficiency simultaneously. Thus, the default m is 5.

4) Influence of the Balance Coefficients α and β: The balance coefficients varied within [0.1, 0.3, 0.5, 0.7, 0.9]. The classification results in terms of OA with multiple coefficient couples are provided in Fig. 6. The biggest OA is obtained when $\alpha = 0.5$ and $\beta = 0.5$. Furthermore, when selecting other coefficient couples, the result is also satisfactory, showing that smMIID is robust to α and β. The default α and β are 0.5.

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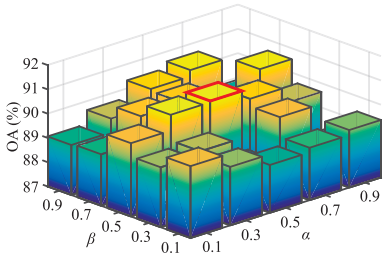


Fig. 6. Classification results in terms of OA (%) with different αs and βs on the T1 subscene of the Houston 2018 dataset.

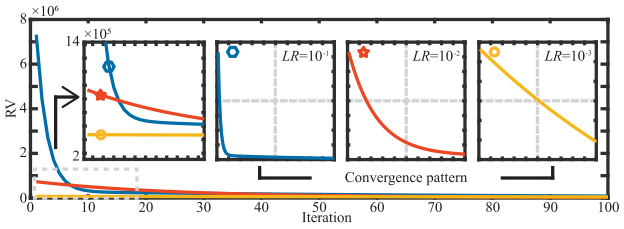


Fig. 7. Convergence curves with different LRs on the T1 subscene.

5) Influence of the Learning Rate LR: The convergence curves with different learning rates LRs within the first 100 iterations are provided in Fig. 7. Specifically, LR was selected within $[10^{-1}, 10^{-2}, 10^{-3}]$. The lines with different colors correspond to the scenarios using different LRs. The initial convergence stages of these curves are enlarged. Furthermore, the convergence patterns (the X-axis and Y-axis values are both normalized into [0, 1]) of the enlarged parts are also provided for capturing their convergence characteristics. The convergence performance was evaluated by the reflectance variance (RV), which is formulated as follows:

$$\mathrm{RV}(\hat{\mathcal{R}}_i) = \sum \|\hat{\mathcal{R}}_i - \hat{\mathcal{R}}_{i-1}\|. \tag{23}$$

Larger LRs always correspond to faster convergence rates and lower RV values. When LR $= 10^{-1}$, the RV value decreases severely and results in too high weight updates. When LR $= 10^{-3}$, the convergence is too slow and results in low updating efficiencies inevitably. In this respect, the default LR is 10^{-2}.

D. Ablation Experiments

Ablation experiments were conducted on the T1 subscene of the Houston 2018 dataset. The classification results of $\mathcal{R}$s, $\mathcal{S}$s and $\mathcal{K}$s in terms of OA in alternative scenarios are shown in Fig. 8. Their descriptions are provided.

1) Contribution of the Shadow Mask $\mathcal{M}^s$: The contribution of the shadow mask $\mathcal{M}^s$ is analyzed initially. Two scenarios are presented as follows.

AS_1: removing $\mathcal{M}^s$ from the constraint on RC.

AS_2: removing $\mathcal{M}^s$ from the constraint on SC.

These scenarios are designed to prove the necessity and effectiveness of introducing $\mathcal{M}^s$ into the energy function. Fig. 8 shows that smMIID outperforms these scenarios and increases the OA of $\mathcal{K}$ by 0.93% and 1.10% when compared with AS_1 and AS_2, respectively. Thus, introducing $\mathcal{M}^s$ in smMIID is feasible and necessary quantitatively.

2) Contribution of the LiDAR Data $\mathcal{L}$: To prove the contribution of $\mathcal{L}$, one scenario is presented as follows.

AS_3: removing $\mathcal{L}$.

The LiDAR data-related constraints are also removed from smMIID. Fig. 8 shows that smMIID outperforms AS_3 in terms of OA. Notably, the OA of $\mathcal{S}$ increases from 74.44% to 88.48% by introducing $\mathcal{L}$ since all constraints on SC are removed from AS_3. Consequently, the OA of $\mathcal{K}$ in smMIID is bigger than that in AS_3, meaning that fusing $\mathcal{R}$ and $\mathcal{S}$ for $\mathcal{K}$ could increase the final performance. These results further verify the necessity of fusing HS and LiDAR data for classification.

3) Contribution of the Hybrid Gradient-Based Correlation: The contribution of the hybrid gradient-based correlation is analyzed by presenting four scenarios as follows.

AS_4: removing the HS gradient-based correlation from the constraints on RC.

AS_5: removing the LiDAR gradient-based correlation from the constraints on RC.

AS_6: removing the hybrid gradient-based correlation from the constraints on RC.

AS_7: removing the LiDAR gradient-based correlation from the constraints on SC.

Fig. 8 shows that smMIID outperforms these scenarios, proving the necessity of considering the hybrid gradient-based correlation during the decomposition process. The OA would decrease if either type of gradient is removed. Furthermore, AS_6 performs worse than both AS_4 and AS_5, illustrating that removing both kinds of gradients is not feasible for smMIID.

4) Contribution of the CIS: The contribution of the CIS is analyzed by presenting one scenario as follows.

AS_8: setting the initial $\hat{\mathcal{R}}$ and $\hat{\mathcal{S}}$ as $\hat{\mathcal{I}}$ and $0_{n\times d}$, respectively.

Fig. 8 clearly proves that this strategy is beneficial to increasing the OA and generating better components. The OA of $\mathcal{K}$ is increased by 0.88% when using the CIS in smMIID.

E. Robustness of smMIID to the Shadow Mask Generation

To intuitively prove the robustness of smMIID to shadow mask generation, multiple generation methods are chosen, including the ground shadow detection (GSD) [34], multicolor-space-based shadow detection (MSD) [35], quaternion cue-based shadow detection (QCSD) [36], region matching of intelligent computing-based shadow detection (RMSD) [37] and shadow detection with multiexposure fusion (SDMF) [22]. DL-based shadow detection techniques are not considered in this work since there are not a large number of labeled/unlabeled images for training. These shadow masks are shown in Fig. 9(b)–(g). Three effective metrics, including the PR, balance error rate (BER), and intersection-over-union (IOU), are chosen for evaluating the generated shadow mask qualities [38]. The formula of PR is provided in (22). BER and IOU are formatted as

$$\begin{aligned} \mathrm{BER} &= \left(1 - \frac{1}{2}\left(\frac{\mathrm{TP}}{N_p} + \frac{\mathrm{TN}}{N_n}\right)\right) \times 100 \\ \mathrm{IOU} &= \frac{\mathrm{region}_{\cap}}{\mathrm{region}_{\cup}} \end{aligned} \tag{24}$$

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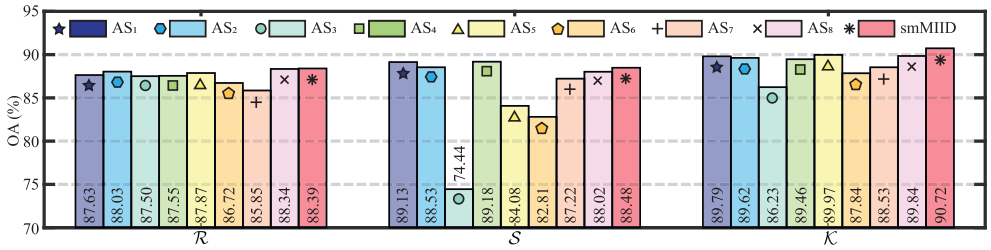


Fig. 8. Classification results in terms of OA (%) with 1% training samples in multiple scenarios on the T1 subscene of the Houston 2018 dataset.

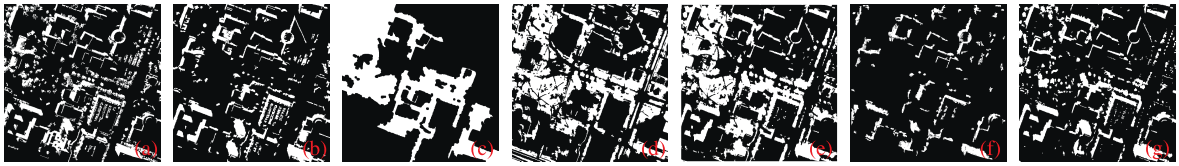


Fig. 9. Shadow masks generated by multiple methods. (a) Shadow reference mask, (b) smMIID, (c) GSD, (d) MSD, (e) QCSD, (f) RMSD, and (g) SDMF.

TABLE VIII

DETECTION RESULTS (DETE), INCLUDING PR (%), BER (%) AND IOU (%), OF DIFFERENT SHADOW MASKS AND CLASSIFICATION RESULTS (CLASS) WITH 1% TRAINING SAMPLES, INCLUDING OA (%), κ ($\times 100$), AND PR (%), USING THESE MASKS ON THE T1 SUBSCENE

	Accuracy	smMIID	GSD	MSD	QCSD	RMSD	SDMF
DETE	PR↑	72.42	61.96	55.41	66.69	**89.53**	75.23
	BER↓	30.71	33.90	42.82	28.15	30.88	**27.75**
	IOU↑	35.54	29.44	20.89	35.87	37.91	**40.21**
CLASS	OA↑	90.72	89.94	90.39	90.07	90.79	**91.03**
	κ↑	78.78	77.23	78.28	77.44	79.06	**79.32**
	PR↑	81.09	75.94	79.86	80.29	80.78	**82.15**

where TP, TN, N_p, and N_n are the numbers of true positive pixels, true negative pixels, shadow pixels, and nonshadow pixels, respectively. $\text{region}_\cap$ and $\text{region}_\cup$ are intersection and union regions. Generally, the smaller the BER value is (↓), the higher the shadow mask quality is. Furthermore, larger PR and IOU values (↑) always correspond to better shadow masks. Notably, the shadow reference mask is manually interpreted using the RGB T1 subscene for evaluation [Fig. 9(a)].

The detection results of the shadow mask generation methods are provided in Table VIII. Furthermore, their corresponding classification results with 1% training samples are also recorded for comparison. Generally, using high-quality shadow masks could generate great classification results. For example, RMSD and SDMF are capable of generating shadow masks with higher detection accuracies than smMIID. Thus, they increase the OA by 0.07% and 0.31% than smMIID, respectively. Fig. 9 shows that GSD cannot generate a satisfactory shadow mask apparently because of the low HS resolution. Its classification accuracies are lower than those of smMIID. However, although these shadow mask qualities are uneven and variable, e.g., the IOU varies within [20.89%, 40.21%], their corresponding classification accuracies are relatively stable, e.g., the OA varies within [89.94%, 91.03%]. These results clearly prove the robustness of smMIID to shadow mask generation. The reason is that smMIID regards the shadow mask as auxiliary information rather than principal information, expanding the application of this technique.

V. CONCLUSION

A novel HS and LiDAR data fusion method called smMIID has been proposed, which is capable of enhancing the HS image quality and strengthening the land-cover classification capability. The motivation is to capture and activate the potential of the crossmodal information in RS modalities and build on the strength of the LiDAR elevation information when constructing HIID constraints. Furthermore, smMIID overcomes the deficiency of information diversity and modality correlation in existing IID-based frameworks. To solve the ill-posed HIID problem, three reasonable constraints on both RC and SC have been constructed simultaneously. Notably, the HS and LiDAR gradients have been comprehensively considered for simulating the local sample correlation. To guarantee spectral consistency, a shadow mask has been utilized when combining neighboring samples. The robustness of smMIID to shadow mask generation has been fully analyzed. Experiments on the widely used datasets have proven the effectiveness of smMIID in terms of visualization and classification. Further experiments will focus on optimizing the energy function.

APPENDIX

The proof of (19) is provided for interested readers. Initially, we introduce some convenient formulas as follows.

Theorem 1: If X is a $n \times d$ matrix, we have

$$\|X\|_F^2 = \text{tr}(X^T X) \tag{25}$$

where $\text{tr}(\cdot)$ is the matrix trace and $(\cdot)^T$ is the matrix transpose.

Theorem 2: If X is a $n \times d$ matrix, M is a $d \times d$ matrix, H is a $n \times d$ matrix, $S = (XM) \odot H$, $f = \|S\|_F^2$, we have

$$\frac{\partial f}{\partial X} = 2[(XH) \odot H \odot H]M^T. \tag{26}$$

Proof: First, some properties are provided as follows.

1) $\text{d}(X \odot Y) = X \odot \text{d}Y + \text{d}X \odot Y$;
2) $\text{d}(XY) = X\text{d}Y + \text{d}X(Y)$;
3) $\text{dtr}(X^T Y) = \text{tr}[(\text{d}X)^T Y + X^T(\text{d}Y)]$;
4) $\text{tr}[X^T(Y \odot Z)] = \text{tr}[(X \odot Y)^T Z]$; and
5) $\text{tr}[(XY)^T Z] = \text{tr}[Y^T(X^T Z)] = \text{tr}[X^T(ZY^T)]$

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where Y and Z have the same shape with X. Thus, we have

$$\begin{aligned} \mathrm{d}S &= \mathrm{d}[(XM)\odot H] = \mathrm{d}(XM)\odot H + (XM)\odot \mathrm{d}H \\ &= \mathrm{d}(XM)\odot H = [(\mathrm{d}X)M + X\mathrm{d}M]\odot H \\ &= [(\mathrm{d}X)M]\odot H \end{aligned} \tag{27}$$

$$\begin{aligned} \mathrm{d}f &= \mathrm{d}\big[\mathrm{tr}(S^T S)\big] = \mathrm{tr}\big[(2S)^T \mathrm{d}S\big] \\ &= \mathrm{tr}\big\{2[(XM)\odot H\odot H]^T(\mathrm{d}X)M\big\} \\ &= \mathrm{tr}\Big\{2\big[((XM)\odot H\odot H)M^T\big]^T \mathrm{d}X\Big\}. \end{aligned} \tag{28}$$

Since $\mathrm{d}f = \mathrm{tr}[(\partial f/\partial X)^T \mathrm{d}X]$, we obtain $(\partial f/\partial X)$ in Theorem 2. ■

For $\mathcal{E}_1(\hat{\mathcal{R}},\hat{\mathcal{S}})$, its partial derivatives are as follows:

$$\begin{aligned} \mathcal{E}_1(\hat{\mathcal{R}},\hat{\mathcal{S}}) &= \mathrm{tr}\Big(\big(\hat{\mathcal{I}}-\hat{\mathcal{R}}-\hat{\mathcal{S}}\big)^T\big(\hat{\mathcal{I}}-\hat{\mathcal{R}}-\hat{\mathcal{S}}\big)\Big) \\ \frac{\partial \mathcal{E}_1}{\partial \hat{\mathcal{R}}} &= \frac{\partial \mathcal{E}_1}{\partial \hat{\mathcal{S}}} = 2\hat{\mathcal{R}}+2\hat{\mathcal{S}}-2\hat{\mathcal{I}}. \end{aligned} \tag{29}$$

For the first element $\mathcal{E}_2^1(\hat{\mathcal{R}}) = \|\hat{\mathcal{R}}\odot\mathcal{M}^s - \mathcal{W}^A\hat{\mathcal{R}}\|_F^2$ in $\mathcal{E}_2(\hat{\mathcal{R}})$, it can be transformed as follows:

$$\begin{aligned} \mathcal{E}_2^1(\hat{\mathcal{R}}) &= \mathcal{E}_{2A}^1(\hat{\mathcal{R}}) - \mathcal{E}_{2B}^1(\hat{\mathcal{R}}) - \mathcal{E}_{2C}^1(\hat{\mathcal{R}}) + \mathcal{E}_{2D}^1(\hat{\mathcal{R}}) \\ \mathcal{E}_{2A}^1(\hat{\mathcal{R}}) &= \mathrm{tr}\Big(\big(\hat{\mathcal{R}}\odot\mathcal{M}^s\big)^T\big(\hat{\mathcal{R}}\odot\mathcal{M}^s\big)\Big) \\ \mathcal{E}_{2B}^1(\hat{\mathcal{R}}) &= \mathrm{tr}\Big(\big(\hat{\mathcal{R}}\odot\mathcal{M}^s\big)^T\big(\mathcal{W}^A\hat{\mathcal{R}}\big)\Big) \\ \mathcal{E}_{2C}^1(\hat{\mathcal{R}}) &= \mathrm{tr}\Big(\big(\mathcal{W}^A\hat{\mathcal{R}}\big)^T\big(\hat{\mathcal{R}}\odot\mathcal{M}^s\big)\Big) \\ \mathcal{E}_{2D}^1(\hat{\mathcal{R}}) &= \mathrm{tr}\Big(\big(\mathcal{W}^A\hat{\mathcal{R}}\big)^T\big(\mathcal{W}^A\hat{\mathcal{R}}\big)\Big). \end{aligned} \tag{30}$$

Inspired by (26), their partial derivatives are as follows:

$$\begin{aligned} \frac{\partial \mathcal{E}_2^1}{\partial \hat{\mathcal{R}}} &= \frac{\partial \mathcal{E}_{2A}^1}{\partial \hat{\mathcal{R}}} - \frac{\partial \mathcal{E}_{2B}^1}{\partial \hat{\mathcal{R}}} - \frac{\partial \mathcal{E}_{2C}^1}{\partial \hat{\mathcal{R}}} + \frac{\partial \mathcal{E}_{2D}^1}{\partial \hat{\mathcal{R}}} \\ \frac{\partial \mathcal{E}_{2A}^1}{\partial \hat{\mathcal{R}}} &= 2(\hat{\mathcal{R}}\odot\mathcal{M}^s\odot\mathcal{M}^s) \\ \frac{\partial \mathcal{E}_{2B}^1}{\partial \hat{\mathcal{R}}} &= \mathcal{M}^s\odot(\mathcal{W}^A\hat{\mathcal{R}}) + (\mathcal{W}^A)^T(\mathcal{M}^s\odot\hat{\mathcal{R}}) \\ \frac{\partial \mathcal{E}_{2C}^1}{\partial \hat{\mathcal{R}}} &= \frac{\partial \mathcal{E}_{2B}^1}{\partial \hat{\mathcal{R}}} \\ \frac{\partial \mathcal{E}_{2D}^1}{\partial \hat{\mathcal{R}}} &= 2(\mathcal{W}^A)^T\mathcal{W}^A\hat{\mathcal{R}}. \end{aligned} \tag{31}$$

Since $\mathcal{E}_2^1(\hat{\mathcal{R}})$ and the second element $\mathcal{E}_2^2(\hat{\mathcal{R}})$ in $\mathcal{E}_2(\hat{\mathcal{R}})$ are consistent formally, we get the $\mathcal{E}_2(\hat{\mathcal{R}})$ partial derivatives $(\partial\mathcal{E}_2/\partial\hat{\mathcal{R}})$ and $(\partial\mathcal{E}_2/\partial\hat{\mathcal{S}})$ (equals to 0). Similarily, we get the $\mathcal{E}_3(\hat{\mathcal{S}})$ partial derivatives, including $(\partial\mathcal{E}_3/\partial\hat{\mathcal{R}})$ (equals to 0) and $(\partial\mathcal{E}_3/\partial\hat{\mathcal{S}})$. By combining these partial derivatives, we get $(\partial\mathcal{E}/\partial\hat{\mathcal{R}})$ and $(\partial\mathcal{E}/\partial\hat{\mathcal{S}})$ in (19).

REFERENCES

[1] Y. Gu and Q. Wang, "Discriminative graph-based fusion of HSI and LiDAR data for urban area classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 6, pp. 906–910, Jun. 2017.

[2] Y. Yang, D. Zhu, T. Qu, Q. Wang, F. Ren, and C. Cheng, "Single-stream CNN with learnable architecture for multisource remote sensing data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5409218.

[3] M. D. Mura, J. A. Benediktsson, B. Waske, and L. Bruzzone, "Morphological attribute profiles for the analysis of very high resolution images," *IEEE Trans. Geosci. Remote Sens.*, vol. 48, no. 10, pp. 3747–3762, Oct. 2010.

[4] W. Dong, T. Zhang, J. Qu, S. Xiao, T. Zhang, and Y. Li, "Multibranch feature fusion network with self- and cross-guided attention for hyperspectral and LiDAR classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5530612.

[5] Z. Xue, X. Tan, X. Yu, B. Liu, A. Yu, and P. Zhang, "Deep hierarchical vision transformer for hyperspectral and LiDAR data classification," *IEEE Trans. Image Process.*, vol. 31, pp. 3095–3110, 2022.

[6] H. Fu, G. Sun, A. Zhang, B. Shao, J. Ren, and X. Jia, "Tensor singular spectral analysis for 3D feature extraction in hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5403914.

[7] H. Fu, G. Sun, J. Ren, A. Zhang, and X. Jia, "Fusion of PCA and segmented-PCA domain multiscale 2-D-SSA for effective spectral–spatial feature extraction and data classification in hyperspectral imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5500214.

[8] Y. Peng, Y. Zhang, B. Tu, C. Zhou, and Q. Li, "Multiview hierarchical network for hyperspectral and LiDAR data classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1454–1469, 2022.

[9] C. Li, R. Hang, and B. Rasti, "EMFNet: Enhanced multisource fusion network for land cover classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 4381–4389, 2021.

[10] S.-B. Chen, Q.-S. Wei, W.-Z. Wang, J. Tang, B. Luo, and Z.-Y. Wang, "Remote sensing scene classification via multi-branch local attention network," *IEEE Trans. Image Process.*, vol. 31, pp. 99–109, 2022.

[11] D. Hong, L. Gao, R. Hang, B. Zhang, and J. Chanussot, "Deep encoder–decoder networks for classification of hyperspectral and LiDAR data," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[12] M. Zhang, W. Li, R. Tao, H. Li, and Q. Du, "Information fusion for classification of hyperspectral and LiDAR data using IP-CNN," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506812.

[13] M. Zhang, W. Li, Y. Zhang, R. Tao, and Q. Du, "Hyperspectral and LiDAR data classification based on structural optimization transmission," *IEEE Trans. Cybern.*, vol. 53, no. 5, pp. 3153–3164, May 2023.

[14] F. Zhang, X. Jiang, Z. Xia, M. Gabbouj, J. Peng, and X. Feng, "Nonlocal color compensation network for intrinsic image decomposition," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 33, no. 1, pp. 132–145, Jan. 2023.

[15] Q. Zhang, J. Zhou, L. Zhu, W. Sun, C. Xiao, and W.-S. Zheng, "Unsupervised intrinsic image decomposition using internal self-similarity cues," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 12, pp. 9669–9686, Dec. 2022.

[16] D. Forsyth and J. J. Rock, "Intrinsic image decomposition using paradigms," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 11, pp. 7624–7637, Nov. 2022.

[17] H. Zhang and J. Ma, "IID-MEF: A multi-exposure fusion network based on intrinsic image decomposition," *Inf. Fusion*, vol. 95, pp. 326–340, Jul. 2023.

[18] G. Fu, Q. Zhang, and C. Xiao, "Towards high-quality intrinsic images in the wild," in *Proc. IEEE Int. Conf. Multimedia Expo (ICME)*, Jul. 2019, pp. 175–180.

[19] Y. Gu, W. Xie, X. Li, and X. Jin, "Hyperspectral intrinsic image decomposition with enhanced spatial information," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5523014.

[20] X. Kang, S. Li, L. Fang, and J. A. Benediktsson, "Intrinsic image decomposition for feature extraction of hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 53, no. 4, pp. 2241–2253, Apr. 2015.

[21] X. Jin, Y. Gu, and T. Liu, "Intrinsic image recovery from remote sensing hyperspectral images," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 224–238, Jan. 2019.

[22] P. Duan, S. Hu, X. Kang, and S. Li, "Shadow removal of hyperspectral remote sensing images with multiexposure fusion," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537211.

[23] A. S. Baslamisli, Y. Liu, S. Karaoglu, and T. Gevers, "Physics-based shading reconstruction for intrinsic image decomposition," *Comput. Vis. Image Understand.*, vol. 205, Apr. 2021, Art. no. 103183.

[24] Y. Liu, Y. Li, S. You, and F. Lu, "Unsupervised learning for intrinsic image decomposition from a single image," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3245–3254.

[25] G. Gao, B. Liu, X. Zhang, X. Jin, and Y. Gu, "Multitemporal intrinsic image decomposition with temporal–spatial energy constraints for remote sensing image analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5403116.

[26] X. Xu, W. Li, Q. Ran, Q. Du, L. Gao, and B. Zhang, "Multisource remote sensing data classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 937–949, Feb. 2018.

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[27] R. Hang, Z. Li, P. Ghamisi, D. Hong, G. Xia, and Q. Liu, "Classification of hyperspectral and LiDAR data using coupled CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 7, pp. 4939–4950, Jul. 2020.
[28] X. Kang, Y. Huang, S. Li, H. Lin, and J. A. Benediktsson, "Extended random for shadow detection in very high resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 867–876, Feb. 2018.
[29] E. H. Land, "The retinex theory of color vision," *Sci. Amer.*, vol. 237, no. 6, pp. 108–129, 1977.
[30] W. Yu, M. Zhang, and H. Huang, "Accelerated adaptive feature balance technique based on TEMD for hyperspectral classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.
[31] S. Mohla, S. Pande, B. Banerjee, and S. Chaudhuri, "FusAtNet: Dual attention based SpectroSpatial multimodal fusion network for hyperspectral and LiDAR classification," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 416–425.
[32] W. Yu, H. Huang, and G. Shen, "Deep spectral–spatial feature fusion-based multiscale adaptable attention network for hyperspectral feature extraction," *IEEE Trans. Instrum. Meas.*, vol. 72, pp. 1–13, 2023.
[33] W. Yu, M. Zhang, Z. He, and Y. Shen, "Convolutional two-stream generative adversarial network-based hyperspectral feature extraction," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5506010.
[34] J. F. Lalonde, A. A. Efros, and S. G. Narasimhan, "Detecting ground shadows in outdoor consumer photographs," in *Proc. Eur. Conf. Comput. Vis.*, 2010, pp. 322–335.
[35] A. Martin, M. N. Kamaruddin, and Z. M. Sani, "Shadow detection and removal of road images with maximum extraction of road marker using multi-colour-space," in *Proc. Int. Symp. Intell. Signal Process. Commun. Syst. (ISPACS)*, Nov. 2022, pp. 1–4.
[36] H. Hiary, R. Zaghloul, and M. B. Al-Zoubi, "Single-image shadow detection using quaternion cues," *Comput. J.*, vol. 61, no. 3, pp. 459–468, Mar. 2018.
[37] J. Feng, Y. K. Kim, and P. Liu, "Image shadow detection and removal based on region matching of intelligent computing," *Comput. Intell. Neurosci.*, vol. 2022, pp. 1–9, Apr. 2022.
[38] L. Jie and H. Zhang, "RMLANet: Random multi-level attention network for shadow detection and removal," *IEEE Trans. Circuits Syst. Video Technol.*, early access, Jun. 6, 2023, doi: 10.1109/TCSVT.2023.3283416.



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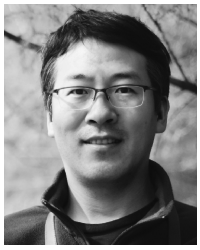
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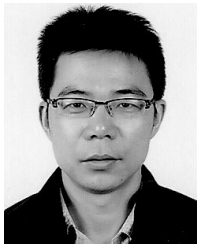
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